

THE ELEMENTS OF FINANCE

A Plain Introduction for First-Year Students



MONEY · TIME · RISK · VALUE

With Worked Cases, Short Answers, Glossary and Index

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A Plain Introduction for First-Year Students

Money, Time, Risk and Value, taught with worked cases from a small Bangladeshi firm

FIRST EDITION
2026

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Spelling follows British usage. Money is in Bangladeshi taka (Tk.), written in lakh and crore grouping (1,00,000 = one lakh). Tax rates and market rates in the worked cases are assumptions chosen for teaching; they are not current law or market quotes.

Karnaphuli Packaging Ltd is an illustrative business created for this book. It is not a real company, and no resemblance to any real firm is intended.

Set in Caladea. Printed in black ink on cream uncoated paper.

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TO THE READER

This book assumes that you know no finance at all. It does not assume that you are slow. You can already do arithmetic and a little algebra, and that is enough. What is new is the vocabulary and the ideas, so the book goes at an even pace and says each thing plainly.

Look at the sample page below. The **wide column** carries the argument. The **narrow column** at the outer edge carries notes. An *italic note* gives the paragraph in one line, so you can find your place again. A note with a **bold word** gives a plain definition of a new term.

A new term is printed in **bold** the first time it appears, and is explained at once. After that it appears in ordinary type. Italics are used for emphasis and titles. Every term is collected in the Glossary, and the Index shows every page where it is used.

Where an idea is difficult, a note says *Go slowly here*, and the passage is taken one step at a time. Calculations are set in a framed box in the same order every time: Given, Find, Formula, Steps, Answer, Check. Read the Check line; it teaches you to test any answer yourself.

Each section ends with a short recap, and each chapter with a summary, a problem to think through, common mistakes, and a true story from history. Review questions have short answers at the back. The tables of factors and a page of basic mathematics are also at the back.



Italic note: the paragraph in one line

Bold term: a plain definition

Bold word in text: a new term, explained at once

Main column: the argument and worked examples

PREFACE

Money is the most ordinary subject in the world and the most poorly taught. People handle it every day, yet when they meet it in a lecture hall it arrives in a shower of symbols. This book tries to restore the order: first the everyday experience, then the idea, then the rule, and last the exceptions.

The book has eight chapters. The first asks what finance is for. The second teaches the value of money through time, on which everything else rests. The third and fourth show how a firm chooses investments and how risk and borrowing change the result. The fifth prices bonds and shares. The sixth deals with daily cash. The seventh surveys long-term funds. The eighth considers leasing.

Chapters differ in length because ideas differ in weight. The value of money through time, the choice of investments and the measure of risk are the foundations of the subject, and they receive the most room. Leasing is important but narrow, and it receives the least.

One small business, Karnaphuli Packaging Ltd, runs through the book. It is invented, and its facts are given in a case file so that you may check every figure. Historical stories, by contrast, are real and have been kept short.

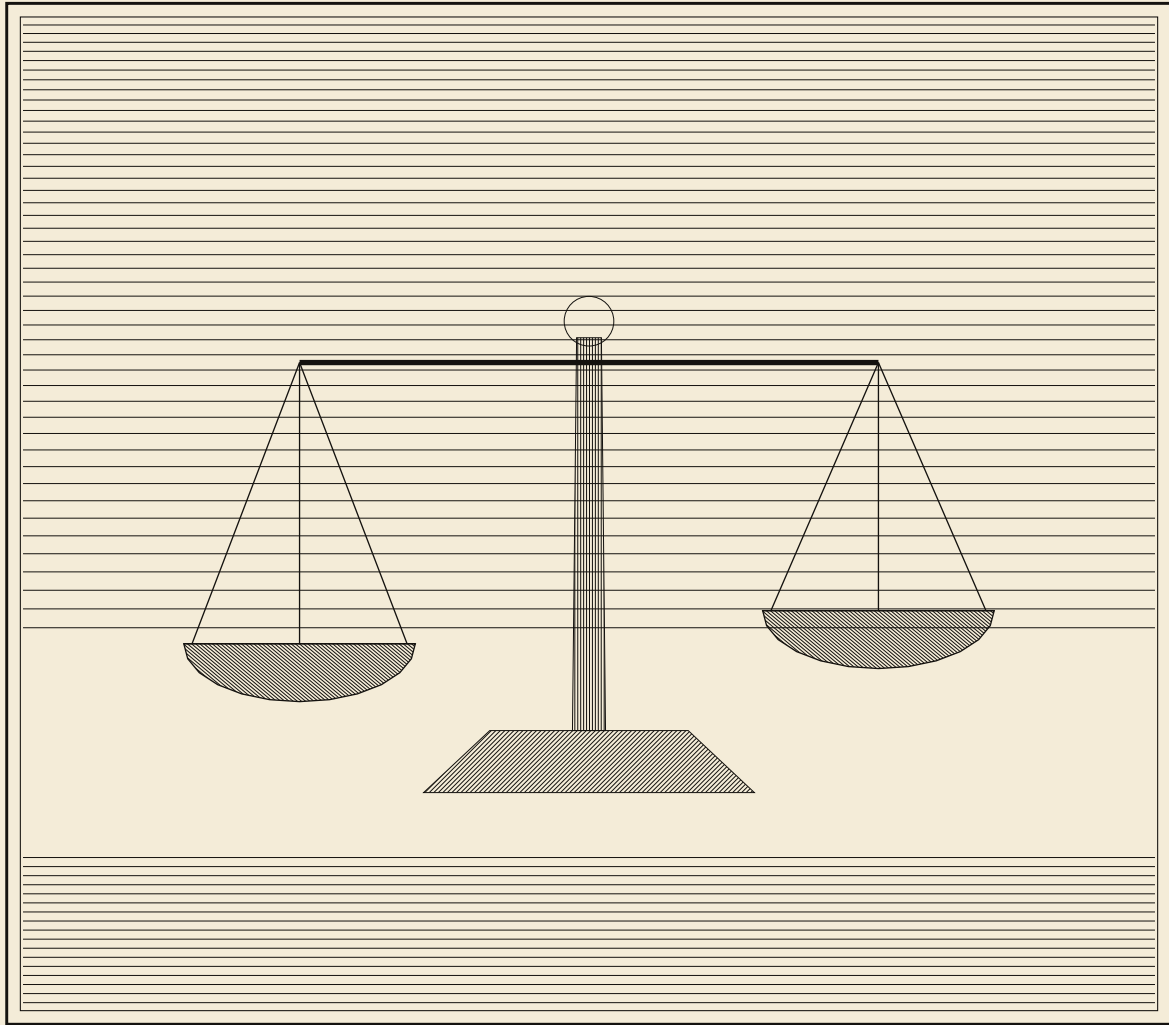
I hope the book leaves you with a habit more than a body of facts: when someone proposes to spend money, ask what it will bring, when, and how sure we are.

THE CASE FILE

Karnaphuli Packaging Ltd (illustrative). Corrugated cartons for small exporters in Chattogram.

Date	Fact or decision	Chapter
1 Jan 2025	Two founders contribute equity of Tk. 20,00,000; machine bought for Tk. 14,00,000; bank loan Tk. 4,00,000; no dividend	1
2026	Founders hire a manager; part of pay linked to profit and cash	1
2026	Reserve fund: Tk. 3,00,000 needed in four years at 8%; yearly saving Tk. 66,576	2
2026	Projects A and B, Tk. 10,00,000 each; required return 12%; Project A chosen (NPV Tk. 1,96,832)	3
2026 plan	80,000 cartons; price Tk. 50; variable cost Tk. 30; fixed cost Tk. 8,00,000; interest Tk. 1,00,000; break-even 40,000 cartons	4
2027	Plan to issue Tk. 1,000 bonds, 10% coupon, five years	5
2026	Sales Tk. 40,00,000; working capital needed Tk. 8,80,000; supplier terms 2/15, net 45	6
2027	New hall Tk. 40,00,000; retained earnings Tk. 8,00,000; WACC 10.8%	3, 7
2027	Delivery van Tk. 5,00,000: buy (present cost Tk. 3,76,994) or lease (Tk. 4,01,819)	8

Figures in the case file are assumptions chosen to teach each method. They are not forecasts of any real firm. Each chapter that uses the firm repeats the figures it needs.



CHAPTER THE FIRST

What Finance Is For

Who decides what to do with money?



This chapter explains what finance is, the decisions a financial manager makes, the goal that guides those decisions, and why owners and managers sometimes want different things.

A business must answer money questions before it makes anything.

Suppose two friends save Tk. 20,00,000 and open a small factory. On the first morning, three questions stand before them. Which machines should they buy? Where will the rest of the money come from? Who will decide how much profit to keep? None of these questions is about making boxes. All of them are about money. Most businesses succeed or fail on how well such questions are answered.

Finance The study of raising, using and managing money over time.

Finance is household money skills at business size.

Risk The chance that a result differs from what we expected.

Two facts drive finance: time and risk.

Conceptual framework The main ideas that tie a subject together.

Funds flow in, are invested, and flow back.

Firm A business that sells goods or services to earn a profit.

Capital Money that owners and lenders put into a firm.

Word in use: capital is money, not a city.

THE IDEA OF FINANCE

Finance means the study of how people and firms raise money, use it, and manage it over time. Think of a family. It earns an income, decides what to buy, saves a little, and sometimes borrows to pay for a house. A firm does the same things on a larger scale and with more at stake.

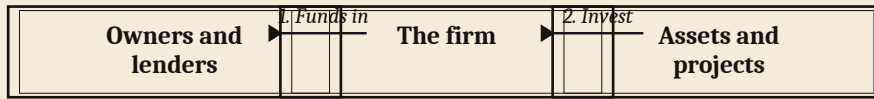
Finance rests on two simple facts. First, money received today is worth more than the same money received later. Second, nobody knows the future for certain, so every plan carries **risk**, which means the chance that the result will differ from what we expect. Almost every idea in this book comes from these two facts: time and risk.

The **conceptual framework** of finance is the set of ideas that tie the subject together. A firm obtains funds from owners and lenders, invests them in assets, earns cash, and returns some cash to those who supplied the funds. Finance studies each step of this circle.

The word *firm* appears often in this book. A **firm** means any business organisation that sells goods or services to earn a profit. We use *firm*, *company* and *business* for the same thing, and we keep to the word *firm*.

WORDS WITH TWO MEANINGS

Capital needs care. In daily life it can mean a city, such as Dhaka. In finance, **capital** means the money that owners and lenders put into a firm so that it can operate.



3. Cash earned returns to the firm, and part goes back to owners and lenders

FIG. 1. The circle of funds in a firm.

BEFORE MOVING ON

Finance is the study of raising, using and managing money. It rests on two facts: money has a time value, and the future is uncertain. A firm is a business that sells to earn a profit. Capital is the money put into it.

THREE DECISIONS

Every financial manager faces three big decisions. The first is the **investment decision**, which means choosing which assets the firm should buy. An asset is anything of value that the firm owns, such as a machine, a stock of paper, or cash in the bank. A good investment earns more than it costs.

The second is the **financing decision**, which means choosing where the money will come from. The firm can ask owners for more money, or it can borrow. Borrowed money must be repaid with interest, so the choice changes the firm's risk.

The third is the **dividend decision**. A *dividend* is a share of profit paid to owners. The firm can pay it out or keep it to grow. Profit that is kept is called **retained earnings**. If the firm keeps more, owners receive less today but may gain more later.

Investment decision Choosing which assets the firm should buy.

Decision one: what to buy.

Financing decision Choosing where the firm's money will come from.

Decision two: where to find the money.

Dividend decision Choosing how much profit to pay out and how much to keep.

Retained earnings Profit that the firm keeps instead of paying out.

Decision three: pay out or keep.

CASE FILE: KARNAPHULI PACKAGING LTD (ILLUSTRATIVE)

Date	1 January 2025. Two founders contribute equity of Tk. 10,00,000 each.
Capital	Total equity Tk. 20,00,000 (twenty lakh; one lakh is 1,00,000).
Decision 1	Investment: buy one corrugating machine for Tk. 14,00,000.
Decision 2	Financing: use all the equity and borrow Tk. 4,00,000 from a bank.
Decision 3	Dividend: pay no dividend in 2025; keep all profit.

Notice how the three decisions connect. The machine (investment) needed more money than the founders had, so they borrowed (financing). The loan left less cash, so they paid no dividend. A change in one decision changes the others.

BEFORE MOVING ON. Name the three decisions: investment, financing, dividend. Retained earnings means profit kept in the firm.

The three decisions are linked, not separate.

Recall the three decisions by name.

Financial manager The person who makes or advises on a firm's money decisions.

Financial planning Estimating how much money a firm will need, and when.

Six functions: plan, raise, allocate, control, protect, report.

Accounting Recording and reporting what a firm has done with its money.

Accounting looks back; finance looks ahead.

Good money management lets every other department work.

Profit maximisation Making accounting profit as large as possible.

Profit ignores time, risk, and accounting tricks.

THE FINANCIAL MANAGER

A **financial manager** is the person who makes or advises on the firm's financial decisions. In a small firm the owner may do this. In a large firm a chief financial officer leads a team. Two roles are common. The *treasurer* looks after cash, borrowing and investing. The *controller* looks after accounting records and reports.

What does the manager actually do? The main functions are these. First, **financial planning**: estimating how much money the firm will need and when. Second, raising funds at fair cost. Third, allocating funds to the best uses. Fourth, controlling results by comparing plans with outcomes. Fifth, protecting the firm against avoidable risks. Sixth, communicating with owners, lenders and tax authorities.

Finance and management are closely tied. A marketing plan needs money for advertising. A production plan needs money for machines. Every plan in a firm ends as a question about money, so the finance function sits near the centre of management. Finance also depends on **accounting**, the record-keeping that tells the manager what has already happened. Accounting records the past; finance uses the record to decide the future.

HOW FINANCE SHAPES SUCCESS

A firm with good financial management buys machines that earn more than they cost. It borrows only what it can repay. It keeps enough cash to pay wages on time. Such a firm runs smoothly and can grow. A firm with poor financial management may be profitable on paper and still fail, because it runs out of cash on the day the wages are due. Operational efficiency, meaning how well the firm turns inputs into output, depends on having the right money in the right place at the right time.

THE GOAL OF THE FIRM

A manager needs a goal, as a driver needs a destination. The simplest goal is **profit maximisation**, which means making the profit as large as possible. It sounds sensible, but it has four weaknesses.

First, it ignores *time*. A profit of Tk. 1,00,000 this year and one of Tk. 1,00,000 in ten years are not equal. Second, it ignores *risk*. A sure Tk. 1,00,000 is better than a doubtful Tk. 1,20,000. Third,

profit can be changed by accounting choices without any real gain. Fourth, it ignores the *timing and amount of cash* that owners receive. A better goal must deal with time and risk together.

That goal is **wealth maximisation**, which means making the value of the owners' shares as large as possible. The value of a share is what buyers will pay for it. It rises when buyers expect large, early, and certain cash in future. In this way one number, the share price, captures size, timing and risk together.

Share value = present value of all future cash that owners expect

EXCEPTIONS. Wealth maximisation is not perfect. Share prices can rise and fall for reasons the firm cannot control. A firm that is not listed on a stock market has no daily price at all. Even so, owners and managers still use wealth as the guide, and they estimate it with the methods taught later.

Wealth maximisation Making the value of the owners' shares as large as possible.

Share value holds size, timing and risk together.

No perfect goal exists, but wealth is the best guide.

OWNERS, MANAGERS AND AGENCY

In a large firm the owners rarely run the business. They hire managers. This creates an **agency relationship**: one person, the **principal**, hires another, the **agent**, to act for them. Owners are principals; managers are agents. Jensen and Meckling described this idea in 1976.

The trouble is that the agent may want something different. A manager may prefer a large office, easy projects, or fast growth that raises his own pay. The owners want share value. When the two goals differ we have an **agency problem**. The cost of the problem is called an **agency cost**. It includes watching the manager, rewarding good behaviour, and the loss that remains.

Agency relationship One person (the principal) hires another (the agent) to act for them.

Owners hire managers; this is the agency link.

Agency problem A clash between what agents want and what owners want.

Agency cost The cost of watching agents, rewarding them, and the loss that remains.

COMMON WAYS TO REDUCE AGENCY PROBLEMS

Method	How it helps
Share options	Manager gains when share price rises
Audit and reports	Owners can see what managers did
Board of directors	A group watches the managers for owners
Lender conditions	Lenders limit risky actions
Threat of takeover	Poor managers may lose their jobs

Karnaphuli Packaging faces a small version of this. In 2026 the founders hire a manager and agree that part of the manager's pay depends on yearly profit and cash. They then check monthly cash reports. These two steps reduce the gap between the founders' wishes and the manager's.

Pay links and checks bring interests closer.

Corporate finance The branch of finance that serves firms' money decisions.

Finance borrows tools from four older subjects.

FINANCE AS A DISCIPLINE

Finance is a young subject that borrows from older ones. From *economics* it takes ideas about choice, cost and markets. From *accounting* it takes data. From *mathematics* it takes the tools for time and risk. From *management* it takes the study of people and decisions. Today finance has three main branches: **corporate finance**, which serves firms; **investments**, which studies securities and markets; and **financial institutions**, which studies banks and similar bodies. This book is mostly about corporate finance.

SUMMARY. Finance studies how money is raised, used and managed under time and risk. The financial manager makes investment, financing and dividend decisions. The best goal is wealth maximisation, not profit alone. Owners (principals) and managers (agents) can disagree, and agency costs measure the damage.

THINK IT THROUGH. A manager can raise this year's profit by cutting machine repairs. What might happen to the firm's value in three years, and who gains today?

WHERE PEOPLE GO WRONG. Many students think that profit and cash are the same. They are not: profit is a record of earning, while cash is money in hand. A firm can have profit and no cash. Another common mistake is to call a dividend an expense; it is a share of profit given to owners.

IN HISTORY. In 1602 the Dutch East India Company, the VOC, sold shares to the public in Amsterdam. Owners could now sell their shares to others without stopping the business. Because investors and managers were now different people, owners had to learn how to watch managers from afar, long before the term *agency cost* existed.

REVIEW QUESTIONS. (1) Define finance in one sentence. (2) Why is wealth maximisation a better goal than profit maximisation? (3) Give one cost of agency and one way to reduce it.



CHAPTER THE SECOND

Money Across Time

Is Tk. 100 today worth the same as Tk. 100 next year?



This chapter shows how to move money through time. It teaches compounding, discounting, annuities and mixed streams, with every step worked by hand.

Sooner money is better; finance measures how much.

Imagine a friend offers you Tk. 1,00,000 today or Tk. 1,00,000 one year from today. You would take the money today, and you would be right. But why? You could put it in a bank and earn interest. You might need it. The future is uncertain. Even a small child understands that sooner is better. Finance puts a number on that feeling.

THE BASIC CONCEPT

Time value of money Money now is worth more than the same money later, because it can earn.

This idea underlies nearly all of finance.

Interest Payment for the use of someone else's money.

Interest rate Interest as a percentage of the sum lent, for one period.

Word in use: interest here is money, not curiosity.

Present value What a future sum is worth today.

Future value What a sum today will be worth at a later date.

Compounding Moving a sum forward in time by adding interest on interest.

Discounting Moving a future sum back to its value today.

Forward is compounding; backward is discounting.

Compound interest Interest earned on both the original sum and earlier interest.

Go slowly here. Interest earns interest.

The **time value of money** means that a sum of money is worth more now than the same sum in the future, because money now can earn a return. It is the most important idea in finance. Without it a firm cannot compare a cost today with a gain in five years.

Two words help us here. **Interest** means the payment for the use of money. If you lend Tk. 100 and receive Tk. 110 a year later, the Tk. 10 is interest. The **interest rate** is the interest as a percentage of the amount lent: here 10 per cent per year.

WORD IN USE. *Present* and *future* have special meanings. The **present value** of a sum is what it is worth today. The **future value** is what it will be worth at a later date. A **period** is one unit of time, usually one year.

Time value has two directions. Moving money *forward* in time is called **compounding**. Moving money *backward* is called **discounting**. The two are mirror images, like climbing a hill and walking down it.

FUTURE VALUES

Go slowly here. Begin with a bank deposit. You place Tk. 100 in an account that pays 10 per cent a year. After one year you have Tk. 110. In the second year interest is paid on Tk. 110, not only on Tk. 100. You earn Tk. 11, and you have Tk. 121. Interest earned on earlier interest is called **compound interest**.

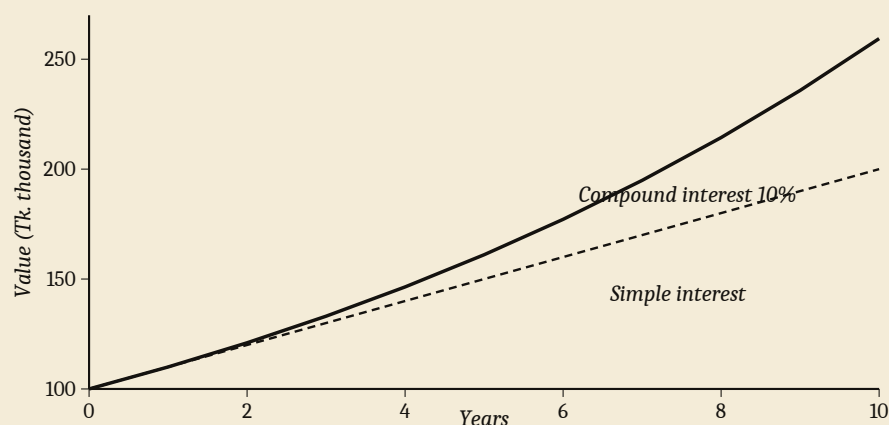


FIG. 2. Tk. 100 grows faster with compound than with simple interest.

Each year the sum is multiplied by 1.10. After n years it is multiplied n times. We turn this pattern into a formula.

$$\text{Future value} = \text{present value} \times (1 + \text{interest rate})^{\text{number of years}}$$

Here FV is future value, PV is present value, i is the interest rate for one period written as a decimal, and n is the number of periods. The factor $(1 + i)^n$ is called the **future value factor**.

Future value factor The number $(1 + i)^n$ that multiplies a present sum to give its future value.

WORKED CASE 1: A SINGLE DEPOSIT	
Given	PV = Tk. 1,00,000; $i = 10\% = 0.10$; $n = 3$ years.
Find	The future value after three years.
Formula	$FV = PV \times (1 + i)^n$
Steps	$(1 + 0.10)^3 = 1.10 \times 1.10 \times 1.10 = 1.331$ $FV = 1,00,000 \times 1.331$
Answer	FV = Tk. 1,33,100.
Check	Year by year: 1,10,000, then 1,21,000, then 1,33,100. The answer agrees.

WORKED CASE 2: SLIGHTLY MESSIER NUMBERS	
Given	Karnaphuli deposits PV = Tk. 2,35,000 at $i = 9\%$ for $n = 4$ years.
Find	The future value.
Formula	$FV = PV \times (1 + i)^n$
Steps	$(1.09)^4 = 1.4116$ (rounded) $FV = 2,35,000 \times 1.4116 = 3,31,726$
Answer	FV is about Tk. 3,31,700.
Check	Simple interest would give $2,35,000 \times (1 + 0.36) = 3,19,600$. Compound interest is higher, as it should be.

COMMON SLIPS. Students often forget the power n and multiply by $(1 + i)$ only once. Others use 10 instead of 0.10. A third slip is mixing periods: if interest is paid every three months, i and n must both be quarterly. For example, Tk. 10,000 at 12% a year paid quarterly for two years uses $i = 3\%$ and $n = 8$, which gives Tk. 12,668.

Match the rate to the period.

Nominal rate The yearly interest rate as quoted, before the effect of compounding.

Effective annual rate The true yearly rate after compounding is counted.

COMPOUNDING MORE THAN ONCE A YEAR

Banks often pay interest quarterly or monthly. The rate quoted for the year is then the **nominal rate**. Because interest earns interest sooner, the real yearly gain is higher. The **effective annual rate** (EAR) is the yearly rate that gives the same result.

Effective annual rate = $(1 + \text{nominal rate} \div m)^m - 1$, where m is the number of periods in a year

WORKED CASE: EFFECTIVE RATE	
Given	Nominal rate 12% a year, compounded quarterly ($m = 4$).
Find	The effective annual rate.
Formula	$\text{EAR} = (1 + i \div m)^m - 1$
Steps	$i \div m = 0.12 \div 4 = 0.03$ $(1.03)^4 = 1.1255$ $\text{EAR} = 1.1255 - 1$
Answer	$\text{EAR} = 12.55\%$ a year.
Check	Annual compounding gives exactly 12%. More frequent compounding gives a little more. Correct.

PRESENT VALUES

Now run the idea backward. What is Tk. 1,33,100, due in three years, worth today if the rate is 10 per cent? Divide by 1.331 and we return to Tk. 1,00,000. Discounting reverses compounding.

Present value = future value $\div (1 + \text{interest rate})^{\text{number of years}}$

The rate used is called the **discount rate**: the rate that converts future money into present money. It reflects what the money could earn elsewhere, and the risk. The bracket is the **present value factor**. Factor tables, such as those at the back of this book, save time.

Discount rate The rate used to turn future sums into present values.

Present value factor The number $1/(1 + i)^n$ that turns a future sum into its present value.

WORKED CASE 3: DISCOUNTING ONE SUM	
Given	FV = Tk. 1,33,100 due in $n = 3$ years; $i = 10\%$.
Find	The present value.
Formula	$\text{PV} = \text{FV} \div (1 + i)^n$
Steps	$(1.10)^3 = 1.331$ $\text{PV} = 1,33,100 \div 1.331 = 1,00,000$
Answer	The present value is Tk. 1,00,000.
Check	Compound Tk. 1,00,000 forward at 10% for three years. We get Tk. 1,33,100.

Self-test sentence for this section.

YOU SHOULD NOW BE ABLE TO SAY. "Compounding multiplies a sum forward; discounting divides a sum back, and the same rate and period must be used in both."

Quick doubling time: 72 divided by the rate.

RULE OF 72. A useful shortcut says that money doubles in about

72 divided by the percentage rate years. At 8%, $72 \div 8 = 9$ years. The rule is approximate. Luca Pacioli recorded a form of it in 1494.

ANNUITIES

Many payments come in regular equal instalments: rent, a loan repayment, a pension. An **annuity** means a series of equal payments made at equal intervals for a fixed number of periods. If we can find one value for the whole series, we save much work.

TYPES OF ANNUITY

Annuities differ by *when* the payment falls in each period. An **ordinary annuity** pays at the *end* of each period; most loans work this way. An **annuity due** pays at the *beginning*; rent paid in advance is an example. A **perpetuity** is an annuity that never ends, such as a fixed payment forever. There is also the **deferred annuity**, which begins after some delay.

THE ANNUITY FAMILY AT A GLANCE

Type	When paid	Example
Ordinary	End of period	Loan instalment
Due	Start of period	Rent in advance
Perpetuity	End, for ever	Fixed payment from a trust
Deferred	After a delay	Pension starting later

Future value of an ordinary annuity = $\text{payment} \times [(1 + i)^n - 1] \div i$

The brackets are the **annuity factors**. An annuity due is worth more than an ordinary annuity, because each payment has one extra period to earn. Multiply the ordinary result by $(1 + i)$ to get the annuity due. A perpetuity's present value equals the payment divided by i .

Annuity A series of equal payments at equal intervals for a fixed time.

Annuity: same amount, same gaps.

Ordinary annuity Equal payments made at the end of each period.

Annuity due Equal payments made at the start of each period.

Perpetuity An annuity that continues for ever.

Annuity factor A number that turns equal payments into one total value.

Due annuities are worth $(1 + i)$ times more.

WORKED CASE 4: FUTURE VALUE OF AN ANNUITY	
Given	Payment = Tk. 20,000 at the end of each year; $i = 8\%$; $n = 5$ years.
Find	The future value at the end of year 5.
Formula	$FV = \text{payment} \times [(1 + i)^n - 1] \div i$
Steps	$(1.08)^5 = 1.4693$ $(1.4693 - 1) \div 0.08 = 5.8666$ $FV = 20,000 \times 5.8666 = 1,17,332$
Answer	The fund will hold about Tk. 1,17,332.
Check	The payments total 1,00,000. The fund is larger because of interest.

WORKED CASE 5: PRESENT VALUE, AND THE DUE CASE	
Given	Payment = Tk. 20,000 yearly; $i = 8\%$; $n = 5$.
Find	Present value (a) ordinary, (b) due.

Formula	$PV = \text{payment} \times [(1 - (1 + i)^{-n}) \div i]$
Steps	$(1.08)^{-5} = 0.6806$ $(1 - 0.6806) \div 0.08 = 3.9927$ (a) $20,000 \times 3.9927 = 79,854$ (b) $79,854 \times 1.08 = 86,243$
Answer	(a) Tk. 79,854; (b) Tk. 86,243.
Check	Five payments of 20,000 add to 1,00,000; each is discounted, so both answers are below that.

Endless payments still have a finite value.

PERPETUITY. If a trust pays Tk. 5,000 every year for ever and the rate is 10 per cent, its present value is $5,000 \div 0.10 = \text{Tk. } 50,000$. This looks surprising, since the payments never stop. The reason is that far-off payments are discounted almost to nothing.

WORKED CASE 5B: A DEFERRED ANNUITY	
Given	Payment Tk. 10,000 at the end of years 3, 4 and 5; $i = 10\%$.
Find	The present value today.
Formula	Value at the start of the payments $\times$ PV factor for the delay
Steps	The three payments form an ordinary annuity that begins after year 2 Annuity factor $(10\%, 3) = 2.4869$; value at end of year 2 = 24,869 Discount back two years: $24,869 \div 1.21 = 20,553$
Answer	The present value is about Tk. 20,553.
Check	The payments total 30,000; discounting for 3 to 5 years brings them to well below that. Sensible.

Self-test for the annuity section.

YOU SHOULD NOW BE ABLE TO SAY. "An annuity is one factor times one payment; a deferred annuity needs one extra step of discounting."

SAVING TO A TARGET

Target $\div$ factor gives the yearly saving.

A common question runs the other way. Karnaphuli needs Tk. 3,00,000 in four years to replace a machine. If savings earn 8 per cent, what must be saved at the end of each year? Divide the target by the annuity factor: $(1.08)^4 = 1.3605$, so the factor is $(1.3605 - 1) \div 0.08 = 4.5061$. The yearly payment is $3,00,000 \div 4.5061 = \text{Tk. } 66,576$.

LOAN INSTALMENTS

Loan $\div$ annuity factor gives the instalment.

A loan is an annuity seen from the lender's side. The amount borrowed is the present value of the instalments. To find the instalment, divide the loan by the annuity factor. Karnaphuli borrows Tk. 4,00,000 at 10 per cent for five years, repaid at each year-end. The factor is $(1 - 1.10^{-5}) \div 0.10 = 3.7908$. The instalment is $4,00,000 \div 3.7908 = \text{Tk. } 1,05,519$.

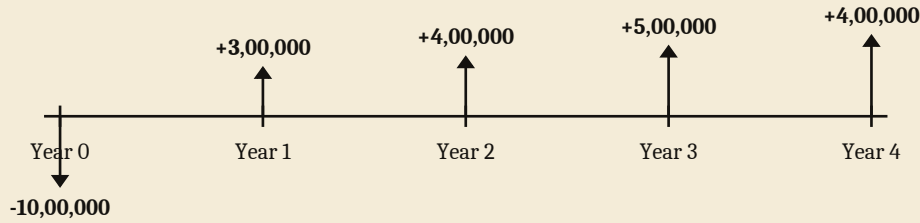


FIG. 3. A time line: cash out at year 0, cash in at years 1 to 4.

MIXED STREAMS

Real cash flows are rarely equal. A **mixed stream** is a series of cash flows whose amounts differ from year to year. We cannot use an annuity factor. Instead we treat each flow as a single sum, discount it separately, and add the results.

Mixed stream A series of cash flows of unequal amounts.

Discount each year alone, then add.

WORKED CASE 6: PRESENT VALUE OF A MIXED STREAM	
Given	Year 1: Tk. 10,000; year 2: Tk. 20,000; year 3: Tk. 30,000; $i = 15\%$.
Find	The present value of the stream.
Formula	$PV = \text{sum of } [\text{cash flow} \div (1 + i)^t] \text{ for each year } t$
Steps	Year 1: $10,000 \div 1.15 = 8,696$ Year 2: $20,000 \div 1.3225 = 15,123$ Year 3: $30,000 \div 1.5209 = 19,725$ Sum = $8,696 + 15,123 + 19,725 = 43,544$
Answer	The present value is about Tk. 43,544.
Check	The cash flows add to 60,000; discounting reduces them. Our answer is lower, as it should be.

To find the *future value* of the same stream we compound each flow to year 3: $10,000 \times 1.3225 + 20,000 \times 1.15 + 30,000 = 13,225 + 23,000 + 30,000 = \text{Tk. } 66,225$. Notice that $PV \times (1.15)^3 = 43,544 \times 1.5209 = 66,225$. The two answers describe the same stream at two dates.

PV and FV are the same stream at two dates.

COMPARING STREAMS. When two streams are discounted at the same rate, the one with the larger present value is worth more today, even if the other has a larger total. Early cash is worth more, so a stream that pays sooner can beat one that pays more later.

Compare streams by present value, not totals.

PRACTICE. (1) Find the future value of Tk. 50,000 at 12% for 5 years. (2) Find the present value of Tk. 10,000 a year for 6 years at 10%. (3) Find the yearly instalment on a Tk. 4,00,000 loan at 10% for 5 years.

Answers are at the back of the book.

SUMMARY. Money has a time value because it can earn. Compounding moves money forward; discounting moves it back. An annuity has equal payments; a mixed stream has unequal ones. Always match the rate and the period.

THINK IT THROUGH. Which is worth more at 12%: Tk. 50,000 now, or Tk. 15,000 at the end of each year for four years?

WHERE PEOPLE GO WRONG. People often use the wrong factor, such as a future value factor to discount. Others forget that an annuity due pays at the start. A good habit is to draw a time line first, mark each payment, and then choose the factor.

IN HISTORY. In 1202 Leonardo of Pisa, called Fibonacci, published *Liber Abaci*. It carried the Hindu-Arabic numerals to European merchants and included problems on profit and interest. With easier numbers, merchants could compare payments made at different dates for the first time.

REVIEW QUESTIONS. (1) Explain time value of money in your own words. (2) How does an annuity due differ from an ordinary annuity? (3) Find the present value of Tk. 5,000 due in two years at 10%.



CHAPTER THE THIRD

Choosing Where to Invest

Which machine should the firm buy, and how do we know?



This chapter shows how a firm chooses long-lasting investments. It follows the steps of capital budgeting, measures the cash a project brings, and compares four evaluation methods.

*One costly choice, hard to undo:
how to decide?*

In 2026 Karnaphuli Packaging can afford only one new project. Project A is a second corrugating line. Project B is a printing unit that lets the firm sell finished, printed cartons. Each costs Tk. 10,00,000. Each will bring cash for four years. A wrong choice will tie up the firm's money for years and cannot easily be undone. How should the founders decide?

Capital budgeting Planning and choosing long-term investments.

Capital expenditure Money spent on assets that bring benefits for many years.

Large, long, hard to reverse, and shape the firm.

Independent projects Projects that can all be accepted if each is good.

Mutually exclusive projects Rival projects: accepting one means rejecting the others.

Relevant cash flow Cash that changes if the project is accepted.

Sunk cost Money already spent that cannot be recovered.

Count future changes only; ignore sunk costs.

Give-up value is also a cost.

BASIC CONCEPTS

Capital budgeting means the process of planning and choosing long-term investments whose benefits last for more than one year. A long-term investment is called a **capital expenditure**. Buying a machine, building a factory, or launching a new product are all examples. Paying this month's electricity bill is not.

Why is capital budgeting necessary? There are four reasons. First, the sums are large compared with the firm's size. Second, the decision lasts a long time, so mistakes are costly. Third, the choice is hard to reverse; a special machine can rarely be resold at its price. Fourth, these decisions set the future size and character of the firm. For an investor the same reasoning applies: each new project changes the value of the shares.

Projects relate to each other in two ways. **Independent projects** can all be accepted if each is good. **Mutually exclusive projects** are rivals: choosing one means rejecting the others. Projects A and B are mutually exclusive, because Karnaphuli can fund only one.

WHICH MONEY COUNTS?

Only cash that changes because of the project belongs in the analysis. This is called **relevant cash flow**. Money already spent and gone is a **sunk cost**; it is the same whether or not we proceed, so it is ignored. Suppose Karnaphuli paid Tk. 20,000 last month for a market study. That money is gone either way and plays no part in the decision.

We also count *opportunity cost*, which means the value of the best alternative given up. If the project uses a room that could be rented for Tk. 30,000 a year, that rent is a cost of the project even though no cash leaves the firm.

THE CAPITAL BUDGETING PROCESS

The process has six steps, and each step needs the one before it.

STEPS OF CAPITAL BUDGETING

Step	What happens
1. Find ideas	Staff, customers and managers suggest projects
2. Estimate cash flows	Forecast costs and benefits for each year
3. Evaluate	Apply measures such as payback and NPV
4. Select	Choose projects using the rules and the budget
5. Implement	Buy, build and start the project
6. Review	Compare actual results with the forecast

Step 2 is the hardest, because forecasts can be wrong. Step 6 is often skipped, yet it teaches the firm how good its forecasts are. When the budget is limited and many projects pass the tests, the firm practises **capital rationing**: it ranks projects and funds the best until the money is used.

Capital rationing Ranking good projects and funding the best until money runs out.

Forecasting is hardest; reviewing teaches most.

CASH FLOW FROM AN INVESTMENT

A project has cash flows at three stages. The **initial investment** is the cash paid at the start: the machine, delivery, installation, and any extra working capital. **Operating cash flows** are the yearly cash benefits during the project's life. The **terminal cash flow** is cash at the end, such as the sale of the machine as scrap, less any closing costs.

Initial investment Cash paid at the start of a project.

Operating cash flow Cash a project brings in each year of its life.

Terminal cash flow Cash received or paid when the project ends.

We use cash flows and not accounting profit, because the firm pays wages and buys materials with cash. One non-cash item still matters: **depreciation**, which means the part of an asset's cost charged as an expense each year. It is not a cash payment. But it reduces taxable profit, so it saves tax. That saving is the *depreciation tax shield*.

Depreciation The share of an asset's cost charged as an expense each year.

Depreciation is not cash, but it cuts tax.

$$\text{Operating cash flow after tax} = (\text{sales} - \text{cash costs} - \text{depreciation}) \times (1 - \text{tax rate}) + \text{depreciation}$$

Read the formula in two parts. The first part is after-tax profit. The second part adds depreciation back, because depreciation never left the firm as cash. We assume a tax rate of 30 per cent in this book's cases; real rates differ by country, industry and year, so check the current law before using one.

WORKED CASE 1: AFTER-TAX OPERATING CASH FLOW	
Given	Sales Tk. 30,00,000; cash costs Tk. 18,00,000; depreciation Tk. 4,00,000; tax rate 30%.
Find	The yearly operating cash flow after tax.
Formula	$\text{OCF} = (\text{sales} - \text{cash costs} - \text{depreciation}) \times (1 - t) + \text{depreciation}$

Steps	Profit before tax = 30,00,000 – 18,00,000 – 4,00,000 = 8,00,000 Tax-paid profit = 8,00,000 × 0.70 = 5,60,000 Add back depreciation: 5,60,000 + 4,00,000 = 9,60,000
Answer	OCF = Tk. 9,60,000.
Check	Without depreciation the tax would be higher. Another way: $12,00,000 \times 0.70 + 4,00,000 \times 0.30 = 8,40,000 + 1,20,000 = 9,60,000$. It agrees.

EVALUATION TECHNIQUES

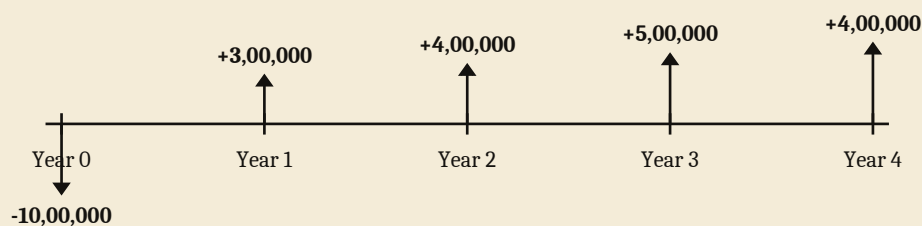


FIG. 4. Cash flows of Project A, in taka.

Here are the forecast cash flows for the two projects. Each costs Tk. 10,00,000 at time zero. The firm's required return is 12 per cent.

FORECAST CASH INFLOWS (Tk.)

Year	Project A	Project B
1	3,00,000	6,00,000
2	4,00,000	4,00,000
3	5,00,000	3,00,000
4	4,00,000	1,00,000
Total	16,00,000	14,00,000

PAYBACK PERIOD

Payback period Time needed for a project's inflows to repay its cost.

Payback asks how soon the money returns.

The **payback period** means the time needed for a project's cash inflows to repay its initial cost. It answers a simple question: how soon do we get our money back? A shorter period is better, and the firm sets a limit, such as three years.

WORKED CASE 2: PAYBACK PERIOD	
Given	Cost Tk. 10,00,000; inflows as in the table.
Find	The payback period of each project.
Formula	Years before full recovery + (unrecovered amount ÷ inflow in recovery year)
Steps	A: after year 2 recovered 7,00,000; left 3,00,000; year 3 inflow 5,00,000; $3,00,000 \div 5,00,000 = 0.6$ B: after year 1 recovered 6,00,000; after year 2 recovered 10,00,000 exactly
Answer	A pays back in 2.6 years; B in 2.0 years.
Check	A: $3,00,000 + 4,00,000 + 0.6 \times 5,00,000 = 10,00,000$. Correct.

Payback is easy to understand, but it has two faults. It ignores cash after the payback date; project A's large year-three and year-four inflows count for nothing. It also ignores the time value of money. These faults mean payback can mislead. Firms use it as a quick screen for liquidity and risk, not as the final test.

The **discounted payback period** repairs the second fault. It discounts each inflow before adding. For Project A at 12 per cent, the discounted inflows are 2,67,857, then 3,18,878, then 3,55,890. After three years the total is 9,42,625, which is still below 10,00,000, so the project needs part of year four: $57,375 \div 2,54,207 = 0.23$. The discounted payback is 3.23 years, longer than the 2.6 years found before.

NET PRESENT VALUE

Go slowly here. **Net present value**, written NPV, means the present value of all a project's cash inflows minus its initial cost. It uses the discount rate from Chapter 2, so it respects time value. If NPV is positive, the project adds value for the owners. If it is negative, the project destroys value.

Net present value = sum of [cash inflow in year $t \div (1 + r)^t$] – initial cost

Here $CF(t)$ is the cash flow in year t , r is the required return, and $CF(0)$ is the initial cost. The **decision rule**: accept a project if NPV is greater than zero. When projects are mutually exclusive, choose the one with the highest positive NPV.

Payback ignores later cash and time value.

Discounted payback period
Time for discounted inflows to repay the cost.

Discounting makes payback longer.

Net present value (NPV)
Present value of a project's inflows minus its cost.

Go slowly here. Positive NPV means value is created.

WORKED CASE 3: NPV OF PROJECT A	
Given	Cost 10,00,000; inflows 3,00,000; 4,00,000; 5,00,000; 4,00,000; $r = 12\%$.
Find	NPV of A.
Formula	$NPV = \sum CF(t) \div (1 + r)^t - CF(0)$
Steps	Year 1: $3,00,000 \div 1.12 = 2,67,857$ Year 2: $4,00,000 \div 1.2544 = 3,18,878$ Year 3: $5,00,000 \div 1.4049 = 3,55,890$ Year 4: $4,00,000 \div 1.5735 = 2,54,207$ Sum = 11,96,832 NPV = 11,96,832 – 10,00,000
Answer	NPV of A = Tk. 1,96,832 (positive: accept).
Check	Inflows total 16,00,000; discounting reduces them to 11,96,832, still above the cost.

WORKED CASE 4: NPV OF PROJECT B, AND THE RANKING	
Given	Cost 10,00,000; inflows 6,00,000; 4,00,000; 3,00,000; 1,00,000; $r = 12\%$.
Find	NPV of B, and which project is better.

Formula	$NPV = \sum CF(t) \div (1 + r)^t - CF(0)$
Steps	Year 1: $6,00,000 \div 1.12 = 5,35,714$ Year 2: $4,00,000 \div 1.2544 = 3,18,878$ Year 3: $3,00,000 \div 1.4049 = 2,13,534$ Year 4: $1,00,000 \div 1.5735 = 63,552$ Sum = 11,31,678; NPV = 1,31,678
Answer	NPV of B = Tk. 1,31,678. Both are acceptable; A has the higher NPV.
Check	B's cash comes earlier, which helps, but its total is smaller, and that costs more.

Methods can disagree; NPV counts all the cash.

Here the two methods disagree. Payback favours B (2.0 years against 2.6), but NPV favours A (Tk. 1,96,832 against Tk. 1,31,678). The NPV method counts all the cash and respects time value, so most financial managers trust it for this choice. Karnaphuli chooses Project A.

SUMMARY OF THE FOUR METHODS

Method	Question it answers	Accept if
Payback	How soon is cost recovered?	Within the set limit
NPV	How much value is added?	NPV above zero
PI	Value per taka invested?	PI above 1.0
IRR	What is the project's own yearly return?	IRR above required return

Profitability index Present value of inflows divided by the initial cost.

Internal rate of return (IRR) The discount rate at which a project's NPV is zero.

Use NPV when methods disagree.

Cost of capital The average rate a firm pays its lenders and owners for funds.
A project must beat the cost of money.

PROFITABILITY INDEX AND INTERNAL RATE OF RETURN

The **profitability index** (PI) means the present value of inflows divided by the initial cost. A PI above 1.0 means NPV is positive. For A it is $11,96,832 \div 10,00,000 = 1.20$; for B it is 1.13. PI helps when money is limited, because it shows value per taka invested.

The **internal rate of return** (IRR) is the discount rate at which NPV equals zero. It is the project's own yearly return. We find it by trial: try a rate, compute NPV, and adjust. For A the IRR is about 20.5 per cent; for B about 20.0 per cent. The rule is to accept a project if IRR is higher than the required return. We introduce IRR here only; later study will use it fully.

EXCEPTIONS. All methods depend on forecasts, and forecasts are uncertain. NPV needs a good discount rate; IRR can give several answers when cash flows change sign more than once; payback ignores value. When methods disagree on mutually exclusive projects, NPV is preferred.

THE COST OF CAPITAL

Where does the discount rate come from? From the firm's **cost of**

capital: the average rate the firm must pay its lenders and owners for the money it uses. It sets a minimum: a project must earn at least this rate, or it harms owners. In that sense it is the *hurdle rate* a project must jump.

Each source has its own cost. Debt costs interest, but interest saves tax, so the *after-tax cost of debt* is the rate times $(1 - \text{tax rate})$. Preferred shares cost their fixed dividend divided by price. Ordinary shares cost what owners expect to earn. We average these costs, weighted by how much each source supplies, to find the **weighted average cost of capital, WACC**.

Weighted average cost of capital (WACC) The average cost of all funds, weighted by each source's share.

WORKED CASE 5: WACC	
Given	Equity 40% at 14%; preferred shares 20% at 12%; debt 40% at 10% before tax; tax rate 30%.
Find	WACC.
Formula	$\text{WACC} = \sum (\text{weight} \times \text{after-tax cost})$
Steps	Debt after tax = $10\% \times 0.70 = 7\%$ Equity: $0.40 \times 14\% = 5.6\%$ Preferred: $0.20 \times 12\% = 2.4\%$ Debt: $0.40 \times 7\% = 2.8\%$ Sum = $5.6 + 2.4 + 2.8$
Answer	WACC = 10.8%.
Check	The result lies between the lowest cost (7%) and highest (14%). Sensible.

Karnaphuli rounds this up to 12 per cent for projects with extra risk. That is why the 12 per cent rate appears in this chapter's cases. Later chapters teach how each cost is found.

PRACTICE. Project C costs Tk. 6,00,000 and returns Tk. 2,00,000 at the end of each of 4 years. Required return 10%. Find (1) the payback period, (2) the NPV, (3) your decision.

SUMMARY. Capital budgeting selects long-term projects using relevant cash flows. Payback is fast but weak. NPV and IRR respect time value; NPV is the safer guide. The cost of capital supplies the discount rate.

THINK IT THROUGH. A project costs Tk. 5,00,000 and returns Tk. 2,00,000 for three years at $r = 10\%$. Compute NPV. Should the firm accept?

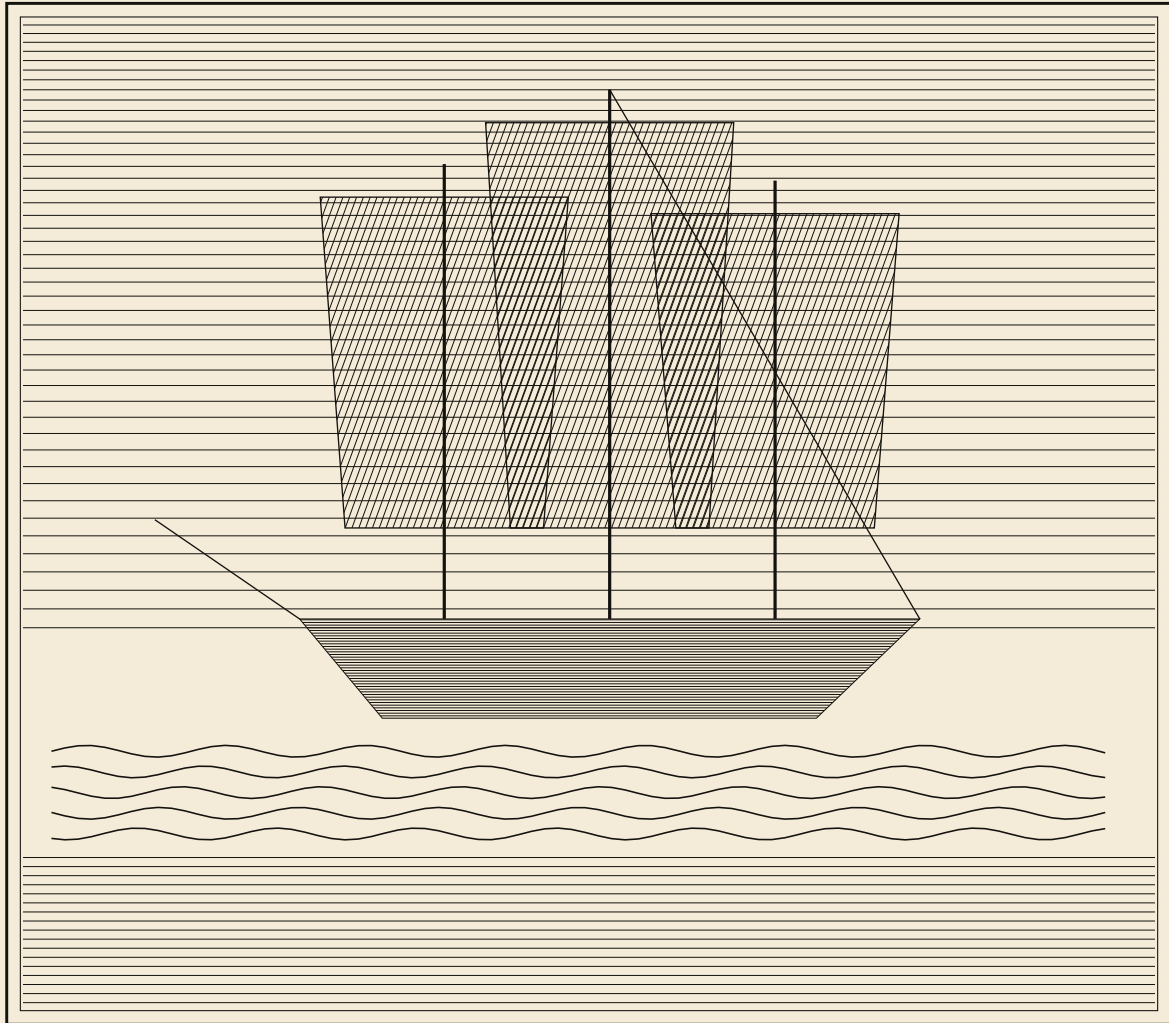
WHERE PEOPLE GO WRONG. Students often include sunk costs, or use accounting profit instead of cash. Others forget to add back depreciation or to subtract the initial cost after discounting. Always write out the three stages: start, years, end.

IN HISTORY. In 1930 the American economist Irving Fisher published *The Theory of Interest*, which set out the logic of valuing future cash by discounting. In 1951 Joel Dean wrote *Capital Budget-*

Answers are at the back of the book.

ing, which brought these methods into the management of firms.

REVIEW QUESTIONS. (1) Why is capital budgeting necessary? (2) Give two faults of the payback method. (3) Define NPV and state its decision rule.



CHAPTER THE FOURTH

Risk, Return and Leverage

Why does a safer firm often earn less?



This chapter explains risk, its kinds and its measures. It shows how fixed costs and borrowing magnify results, how return is priced for risk, and what an efficient market means.

People ask for more reward when risk is higher.

Two friends are offered a choice. One may deposit Tk. 1,00,000 in a bank and receive Tk. 1,06,000 in a year. The other may buy shares that might return Tk. 1,40,000 or might lose half the money. Few people pick the shares unless the possible reward is large. We want to understand this feeling exactly, and measure it.

WHAT RISK IS

Return The gain or loss on an investment, as a percentage of its cost.

Risk means the chance that the result will differ from what we expect. In finance we mean the spread of possible outcomes, good and bad. **Return** means the gain or loss on an investment, usually shown as a percentage of the amount invested. We find a rate of return by dividing the gain by the amount invested.

Rate of return = (income received + change in price) ÷ price at start

If a share bought for Tk. 100 pays Tk. 6 as dividend and is later worth Tk. 104, then $r = (6 + 4) \div 100 = 10$ per cent.

BUSINESS RISK AND FINANCIAL RISK

Business risk Uncertainty about operating income, caused by the business itself.

Financial risk Extra uncertainty caused by the use of borrowed money.

Business risk comes from sales; financial risk from debt.

Risk averse Preferring the safer of two investments with the same expected return.

Business risk means the uncertainty about a firm's operating income, caused by the nature of its business: demand may fall, costs may rise, rivals may arrive. **Financial risk** means the extra uncertainty that comes from borrowing. Interest must be paid in good years and bad, so debt makes owners' returns swing more.

Most people dislike risk. An investor who prefers the surer of two equal returns is **risk averse**. Almost all investors behave this way, and that is why risky assets must promise more.

COMMON KINDS OF RISK

Kind	Meaning
Business	Operating income may vary
Financial	Debt makes owner returns vary more
Market	Events that move all securities together
Interest rate	Rate changes alter prices of bonds
Inflation	Rising prices reduce what money buys
Liquidity	Asset cannot be sold quickly at fair price
Default	Borrower fails to pay as promised

Each kind of risk has a different cause.

Each kind of risk in the table has its own cause. *Market risk* comes from events such as a recession that move all share prices together. *Interest rate risk* means that when rates rise, the prices of existing bonds fall (Chapter 5). *Inflation risk* means that rising prices make the money received in future buy less. *Liquidity risk* means that an asset may be hard to sell quickly without accepting a low price; land is a common case. *Default risk* means that a bor-

rower may fail to pay interest or repay the loan. Lenders reduce default risk by asking for security and by charging higher rates to weaker borrowers.

LEVERAGE

The word *leverage* comes from the lever: a small push at one end moves a heavy load at the other. In finance, **leverage** means the use of fixed costs to magnify the effect of a change in sales on profit. Fixed costs are costs that do not change when output changes, such as rent. Variable costs rise with output, such as paper.

Two kinds of fixed cost create two kinds of leverage. **Operating leverage** comes from fixed *operating* costs, like rent and machines. **Financial leverage** comes from fixed *financing* costs, which means interest on borrowed money. **Total leverage** is the two together.

CASE FILE: KARNAPHULI, PLAN FOR 2026 (ILLUSTRATIVE)	
Units	80,000 cartons.
Price	Tk. 50 per carton. Variable cost Tk. 30 per carton.
Fixed	Fixed operating cost Tk. 8,00,000; interest Tk. 1,00,000.
Income	Sales 40,00,000 – variable cost 24,00,000 = contribution 16,00,000; less fixed 8,00,000 = EBIT 8,00,000.

EBIT means earnings before interest and tax: operating profit. **Contribution** means sales less variable cost; it is what remains to cover fixed costs. The firm breaks even where contribution equals fixed cost: $8,00,000 \div 20$ per carton = 40,000 cartons.

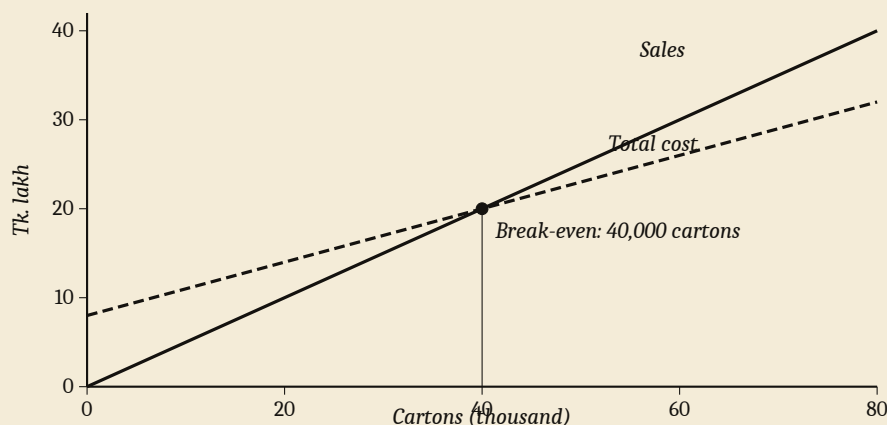


FIG. 5. Sales and total cost lines cross at the break-even point.

DEGREE OF OPERATING LEVERAGE

The **degree of operating leverage** (DOL) means the percentage change in EBIT caused by a one per cent change in sales.

Leverage Using fixed costs to magnify the effect of sales changes on profit.

Fixed costs act as a lever on profit.

Operating leverage Magnified profit change caused by fixed operating costs.

Financial leverage Magnified profit change caused by fixed interest costs.

Total leverage The combined effect of operating and financial leverage.

EBIT Earnings before interest and tax; the profit from operations.

Contribution Sales minus variable cost.

$$DOL = \text{contribution} \div EBIT = Q(P - V) \div [Q(P - V) - F]$$

DOL Percentage change in EBIT for a one per cent change in sales.

Here Q is units, P price, V variable cost per unit, and F fixed operating cost.

WORKED CASE 1: DOL	
Given	Q = 80,000; P = 50; V = 30; F = 8,00,000.
Find	DOL at this output.
Formula	$DOL = Q(P - V) \div [Q(P - V) - F]$
Steps	Contribution = 80,000 × 20 = 16,00,000 EBIT = 16,00,000 – 8,00,000 = 8,00,000 DOL = 16,00,000 ÷ 8,00,000
Answer	DOL = 2.0. A 10% rise in sales lifts EBIT by 20%.
Check	Sales of 88,000 units give contribution 17,60,000 and EBIT 9,60,000, a rise of 20%. Correct.

DEGREE OF FINANCIAL LEVERAGE

DFL Percentage change in earnings after interest for a one per cent change in EBIT.

The **degree of financial leverage** (DFL) means the percentage change in profit after interest, caused by a one per cent change in EBIT.

$$DFL = EBIT \div (EBIT - \text{interest})$$

WORKED CASE 2: DFL AND TOTAL LEVERAGE	
Given	EBIT = 8,00,000; interest = 1,00,000; DOL = 2.0.
Find	DFL and the degree of total leverage, DTL.
Formula	$DFL = EBIT \div (EBIT - I)$; $DTL = DOL \times DFL$
Steps	EBIT – I = 7,00,000 DFL = 8,00,000 ÷ 7,00,000 = 1.143 DTL = 2.0 × 1.143
Answer	DFL = 1.14; DTL = 2.29. A 10% rise in sales lifts earnings after interest by about 22.9%.
Check	EBIT 9,60,000 less interest 1,00,000 = 8,60,000. That is 22.86% above 7,00,000. Correct.

WORKED CASE 2B: A MESSIER CASE	
Given	Price Tk. 120; variable cost Tk. 70; fixed operating cost Tk. 5,00,000; interest Tk. 50,000; Q = 15,000 units.
Find	DOL, DFL and DTL.
Formula	$DOL = \text{contribution} \div EBIT$; $DFL = EBIT \div (EBIT - I)$; $DTL = DOL \times DFL$
Steps	Contribution = 15,000 × 50 = 7,50,000 EBIT = 7,50,000 – 5,00,000 = 2,50,000 DOL = 7,50,000 ÷ 2,50,000 = 3.0 DFL = 2,50,000 ÷ 2,00,000 = 1.25 DTL = 3.0 × 1.25
Answer	DOL = 3.0; DFL = 1.25; DTL = 3.75.
Check	A 10% rise in sales to 16,500 units: EBIT = 8,25,000 – 5,00,000 = 3,25,000, up 30%. Earnings after interest rise from 2,00,000 to 2,75,000, up 37.5%. Correct.

LEVERAGE AND RISK: A COMPARISON OF TWO PLANS

Go slowly here. Suppose Karnaphuli needs Tk. 20,00,000. Plan 1 uses owners' money only. Plan 2 uses Tk. 10,00,000 of owners' money and Tk. 10,00,000 borrowed at 10 per cent, so interest is Tk. 1,00,000. Tax is 30 per cent on profit, with no tax credit on a loss. The measure is the **return on equity**: profit after tax divided by owners' money.

Return on equity Profit after tax divided by the owners' money in the firm.

Go slowly here. Debt raises gains and losses.

RETURN ON EQUITY UNDER THREE LEVELS OF EBIT

EBIT (Tk.)	Plan 1: no debt	Plan 2: with debt
8,00,000	28.0%	49.0%
3,00,000	10.5%	14.0%
80,000	2.8%	-2.0%

WORKED CASE: READING THE TABLE	
Given	Plan 1 equity 20,00,000; Plan 2 equity 10,00,000 and interest 1,00,000; tax 30%.
Find	Return on equity for EBIT 8,00,000 and 80,000.
Formula	$ROE = (EBIT - \text{interest}) \times (1 - t) \div \text{equity}$
Steps	Plan 1, EBIT 8,00,000: $8,00,000 \times 0.70 \div 20,00,000 = 28.0\%$ Plan 2, EBIT 8,00,000: $7,00,000 \times 0.70 \div 10,00,000 = 49.0\%$ Plan 2, EBIT 80,000: $(80,000 - 1,00,000) = -20,000$; no tax; $-20,000 \div 10,00,000 = -2.0\%$
Answer	Debt lifts the return from 28% to 49% in a good year, but turns 2.8% into a loss of 2.0% in a poor year.
Check	The spread of Plan 2 (from -2% to 49%) is wider than that of Plan 1 (2.8% to 28%). More risk, as theory says.

WHY LEVERAGE MATTERS

Leverage works in both directions. If sales fall 10 per cent, earnings after interest fall 22.9 per cent. A firm with high fixed costs gains much in good times and loses much in bad times. So leverage increases risk.

Leverage magnifies both gains and losses.

From the viewpoint of an owner, leverage can raise return. From a lender's viewpoint, it raises the chance of default. For a manager, it narrows the safe margin: below the break-even sales, the firm loses money. A sensible firm chooses its fixed costs with the stability of its sales in mind. Operating and financial leverage together determine the total risk that owners carry.

A steady business can carry more fixed cost.

COMMON SLIPS. Students often use total cost instead of contribution in DOL, or forget that DFL uses EBIT minus interest. If the firm also pays preferred dividends, those are fixed financing costs, too, and the formula grows by a further term.

Probability distribution A list of possible outcomes with the chance of each.

Expected value The average of possible outcomes, weighted by their probabilities.

Standard deviation A measure of how far outcomes spread around their average.

Variance Probability-weighted average of the squared gaps from the expected value.

MEASURING RISK

How do we put a number on risk? We list the possible returns and the *probability* of each, which means its chance of happening, from 0 to 1. The full list is a **probability distribution**. The probabilities must add to 1.

RETURNS ON TWO SECURITIES

State	Probability	Asset A	Asset B
Poor	0.30	4%	-10%
Normal	0.50	10%	12%
Good	0.20	16%	40%

The **expected value** is the average of the outcomes, each weighted by its probability. Multiply each return by its probability and add. For A: $0.30 \times 4 + 0.50 \times 10 + 0.20 \times 16 = 1.2 + 5.0 + 3.2 = 9.4\%$. For B: $-3.0 + 6.0 + 8.0 = 11.0\%$. B offers more on average. But is B better?

WORKED CASE 2C: EXPECTED VALUE OF A SECOND ASSET

Given	Asset X: Poor 0.2 gives 6%; Normal 0.5 gives 10%; Good 0.3 gives 18%.
Find	Expected return, standard deviation and CV of X.
Formula	$E = \sum p \times r$; $SD = \sqrt{\sum p \times (r - E)^2}$; $CV = SD \div E$
Steps	$E = 1.2 + 5.0 + 5.4 = 11.6\%$ Gaps: -5.6, -1.6, +6.4; squares: 31.36, 2.56, 40.96 Weighted: $6.272 + 1.280 + 12.288 = 19.84 = \text{variance}$ $SD = \sqrt{19.84} = 4.45\%$ $CV = 4.45 \div 11.6$
Answer	$E = 11.6\%$; $SD = 4.45\%$; $CV = 0.38$.
Check	X has a higher return than A and a similar spread, so its CV (0.38) is lower than A's (0.45). Consistent.

The **standard deviation** measures how widely outcomes are spread around the expected value. Larger spread means more risk. It is the square root of the **variance**, which is the probability-weighted average of squared gaps from the expected value.

WORKED CASE 3: STANDARD DEVIATION OF A AND B

Given	The table above; $E(A) = 9.4\%$; $E(B) = 11.0\%$.
Find	Standard deviation of each asset.
Formula	$SD = \text{square root of } \sum p \times (r - E)^2$
Steps	$A: 0.30 \times (-5.4)^2 = 8.748$; $0.50 \times (0.6)^2 = 0.180$; $0.20 \times (6.6)^2 = 8.712$; variance 17.64 $SD(A) = \sqrt{17.64} = 4.2\%$ $B: 0.30 \times (-21)^2 = 132.3$; $0.50 \times (1)^2 = 0.5$; $0.20 \times (29)^2 = 168.2$; variance 301 $SD(B) = \sqrt{301} = 17.35\%$
Answer	$SD(A) = 4.2\%$; $SD(B) = 17.35\%$. Asset B is far riskier.

Check B's returns range from -10% to 40%; A's from 4% to 16%. Greater spread, greater SD.

Standard deviation compares well only when expected returns are equal. Here they differ, so we use the **coefficient of variation** (CV): risk per unit of return.

$$CV = \text{standard deviation} \div \text{expected return}$$

$CV(A) = 4.2 \div 9.4 = 0.45$. $CV(B) = 17.35 \div 11.0 = 1.58$. Asset A carries much less risk for each unit of return. A cautious investor prefers A; one who accepts risk for high reward may choose B.

Coefficient of variation

Standard deviation divided by expected return: risk per unit of return.

CV compares risk when returns differ.

RISK OF SEVERAL SECURITIES

Few investors hold one security. A group of securities is a **portfolio**. The expected return of a portfolio is the weighted average of the returns of its parts. If Tk. 60 of every Tk. 100 goes to Asset A (9.4%) and Tk. 40 to Asset B (11.0%), the portfolio return is $0.60 \times 9.4 + 0.40 \times 11.0 = 5.64 + 4.40 = 10.04$ per cent.

Risk behaves differently. Because the assets rarely fall together, the portfolio's risk is usually *less* than the weighted average of the separate risks. This is **diversification**. It removes **unique risk**, which affects one firm only, such as a strike. It cannot remove **market risk**, which affects all firms, such as a recession. Beta measures only market risk, and that is why CAPM pays a premium for it alone.

Portfolio A group of securities held together.

Diversification Spreading money over many securities to reduce risk.

Market risk Risk that affects all securities and cannot be diversified away.

Diversifying removes unique risk, never market risk.

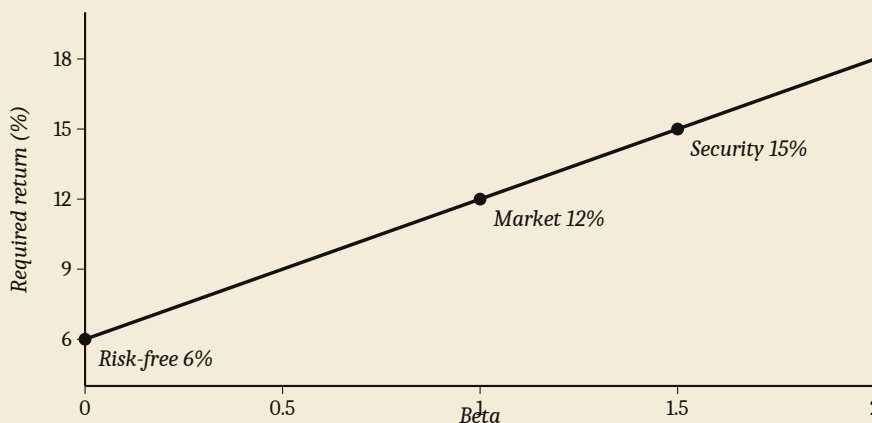


FIG. 6. The required return rises with beta: the security market line.

RISK AND REQUIRED RETURN

Risk-free rate The return on an asset with no chance of default.

Risk premium The extra return demanded for bearing risk.

Required return The minimum return an investor accepts, given the risk.

CAPM A model that sets required return from the risk-free rate, beta and the market premium.

Beta A measure of how much a security moves with the whole market.

An investor needs a reward for taking risk. The **risk-free rate** is the return on an asset with no chance of default, such as a shortterm government security. The **risk premium** is the extra return demanded above the risk-free rate. The **required return** is the sum of the two.

$$\text{Required return} = \text{risk-free rate} + \text{risk premium}$$

The Capital Asset Pricing Model, **CAPM**, sets the premium. It uses **beta**, which measures how strongly a security moves with the whole market. A beta of 1 means it moves with the market; a beta above 1 means it moves more.

$$\text{Required return} = R_f + \text{beta} \times (R_m - R_f)$$

WORKED CASE 4: CAPM

Given	$R_f = 6\%$; market return $R_m = 12\%$; beta = 1.5 (illustrative rates).
Find	The required return and the risk premium.
Formula	$r = R_f + \beta \times (R_m - R_f)$
Steps	Market premium = $12\% - 6\% = 6\%$ $\beta \times \text{premium} = 1.5 \times 6\% = 9\%$ $r = 6\% + 9\%$
Answer	Required return = 15% ; the risk premium on this security = 9% .
Check	Beta above 1 means more risk than the market, so the premium should exceed 6% . It does.

Risk-return trade-off Higher expected return goes with higher risk.

Safe assets pay little; risky assets must pay more.

THE RISK-RETURN TRADE-OFF

Taken together, these ideas give the **risk-return trade-off**: higher expected return goes with higher risk. A market that offered high return with low risk would attract so many buyers that prices would rise and the return would fall. Over time, risky assets must pay more than safe ones, or no one would hold them. For a manager, the lesson is that a project's cost of money rises with its risk.

A portfolio has a beta, too. It is the weighted average of the betas of its parts.

WORKED CASE 4B: BETA OF A PORTFOLIO

Given	Half the money in a security with beta 1.2 and half in one with beta 0.8; $R_f = 6\%$; $R_m = 12\%$.
Find	Portfolio beta and required return.
Formula	Portfolio beta = β (weight $\times$ beta); $r = R_f + \beta \times (R_m - R_f)$
Steps	Beta = $0.50 \times 1.2 + 0.50 \times 0.8 = 0.6 + 0.4 = 1.0$ $r = 6\% + 1.0 \times (12\% - 6\%)$
Answer	Beta = 1.0 ; required return = 12% .
Check	A portfolio with beta 1 moves with the market, so its return should equal the market return. It does.

Answer: P, since the return is the same with less risk.

PAUSE AND CHECK. Two shares have the same expected return,

10%. Share P has SD 5%; share Q has SD 9%. Which would a risk-averse investor choose?

EFFICIENT MARKETS

A market is **efficient** when prices quickly reflect available information. Eugene Fama described three forms in 1970. In the *weak form*, prices reflect all past prices, so chart patterns cannot give profit. In the *semi-strong form*, prices also reflect all public news. In the *strong form*, prices reflect all information, even secret. Most evidence supports the weak form and partly the semi-strong; few believe in the strong form.

PRACTICE. (1) A firm sells 10,000 units at Tk. 90 with variable cost Tk. 60, fixed cost Tk. 1,50,000 and interest Tk. 30,000. Find DOL, DFL and DTL. (2) Find the required return if $R_f = 7\%$, $R_m = 13\%$ and $\beta = 0.8$.

Efficient market A market where prices quickly reflect available information.

Three forms: past prices, public news, all news.

Answers are at the back of the book.

WORKED CASE 5: RETURN ON A SHARE HELD TWO YEARS	
Given	Bought at Tk. 200; dividends of Tk. 10 in total; sold for Tk. 214.
Find	The rate of return over the period.
Formula	$r = (D + P_1 - P_0) \div P_0$
Steps	Price gain = $214 - 200 = 14$ Return in taka = $10 + 14 = 24$ $r = 24 \div 200$
Answer	$r = 12\%$ over two years (about 5.8% a year with compounding).
Check	$(1.058)^2 = 1.119$, close to 1.12. The yearly figure is sensible.

PRACTICE (MORE). (3) Returns on a security are 5% with probability 0.25, 9% with 0.50 and 15% with 0.25. Find the expected return, SD and CV. (4) Compare the ROE of the two financing plans if EBIT is 5,00,000.

Answers are at the back of the book.

SUMMARY. Risk is the spread of outcomes. Business risk comes from operations, financial risk from debt. Leverage magnifies change. Expected value, standard deviation and CV measure risk. Required return equals the risk-free rate plus a premium.

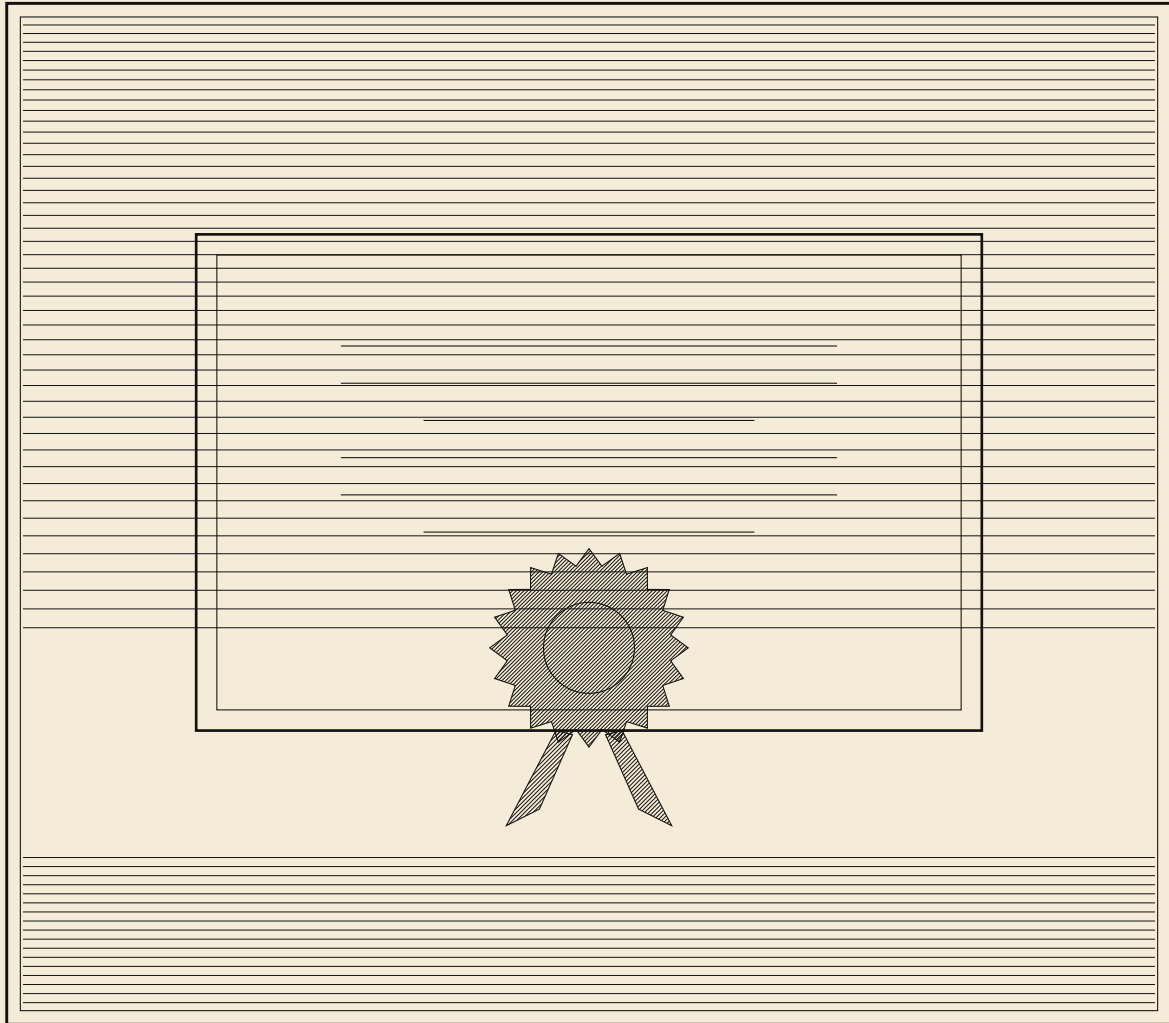
THINK IT THROUGH. Asset X has $E = 12\%$ and $SD = 6\%$. Asset Y has $E = 20\%$ and $SD = 15\%$. Compute each CV. Which is riskier per unit of return?

WHERE PEOPLE GO WRONG. Many students think a high standard deviation means a bad investment; it only means high risk. Others use the CV when expected returns are equal, though SD alone is enough. Remember that leverage cuts both ways.

IN HISTORY. In 1952 Harry Markowitz published *Portfolio Selection*. It showed that the risk of a group of securities depends on how they move together, not only on each one alone. In 1964 Willi-

am Sharpe built on this to develop CAPM.

REVIEW QUESTIONS. (1) Distinguish business risk from financial risk. (2) Explain DOL, DFL and DTL. (3) Calculate the CV of a security with $E = 8\%$ and $SD = 4\%$.



CHAPTER THE FIFTH

What a Security Is Worth

How much should anyone pay for a promise of future cash?



This chapter explains financial securities and teaches how to price two of them: the bond, which promises fixed payments, and the common share, whose payments can grow.

A promise of cash has a price we can compute.

A neighbour offers to sell you a paper that promises Tk. 100 every year for five years, and Tk. 1,000 at the end. What would you pay for it? Pay too much and you lose; ask too little and nobody will sell. The answer comes from two ideas you already know: discounting, and the return that you require.

FUNDAMENTALS OF VALUATION

Security A document giving its holder a claim on future cash.

Bond A loan to an issuer, repaid with fixed payments.

Share A part-ownership of a firm, with a claim on its profit.

Intrinsic value The worth of a security found by discounting its future cash.

Word in use: stock here means ownership.

A **security** means a document that gives its holder a claim on future cash. Two kinds matter most. A **bond** is a loan that the holder makes to the issuer, in return for fixed payments. A **share** (also called *stock*) is a part-ownership of a firm, with a claim on its profit. The firm that sells the security is the *issuer*.

Two words need care. *Stock* in daily trade means goods held for sale; here it means ownership. The **intrinsic value** of a security is its worth based on the cash it will pay, found by discounting. The **market price** is what buyers actually pay today. If intrinsic value is above the market price, the security looks cheap; if below, expensive.

*Value of a security = present value of its expected future cash,
discounted at the required return*

This single idea prices every security. Only the pattern of cash changes: fixed for a bond, uncertain and often growing for a share.

VALUATION OF BONDS

FEATURES OF A BOND

Face value The amount a bond repays at maturity.

Coupon rate Yearly interest on a bond, as a percentage of face value.

Maturity The date when a bond's face value is repaid.

Indenture The legal contract that sets out a bond's terms.

A bond has five main features. The **face value** (or *par value*) is the amount repaid at the end; Tk. 1,000 is common in our examples. The **coupon rate** is the yearly interest, shown as a percentage of face value; a 10 per cent coupon on Tk. 1,000 pays Tk. 100 each year. The **maturity** is the date when the face value is repaid. The **issuer** is the borrower: a firm or the government. The **indenture** is the legal contract that lists the promises and the rights of lenders. Some bonds may be *callable*, which means the issuer may repay them early.

BOND HOLDERS AND SHARE HOLDERS. A bond holder is a lender; a share holder is an owner. The lender is paid first and receives fixed amounts. The owner is paid after the lender and receives what is left, which may be large or nothing.

Lenders are paid first; owners take what is left.

VALUING A BOND

A bond pays two kinds of cash: a yearly coupon, which is an annuity, and the face value at maturity, which is a single sum. We discount both at the required return, using Chapter 2.

$$\text{Bond value} = \text{coupon} \times \text{annuity factor} + \text{face value} \times \text{present value factor}$$

Here I is the yearly interest, k the required return, n the years to maturity, and F the face value.

WORKED CASE 1: BOND WITH A REQUIRED RETURN OF 12%	
Given	$F = \text{Tk. } 1,000$; coupon 10% ($I = \text{Tk. } 100$); $n = 5$ years; $k = 12\%$.
Find	The value of the bond.
Formula	$B = I \times \text{annuity factor} + F \times \text{PV factor}$
Steps	Annuity factor (12%, 5) = 3.6048 PV factor (12%, 5) = 0.5674 Interest part = $100 \times 3.6048 = 360.48$ Principal part = $1,000 \times 0.5674 = 567.43$
Answer	$B = 360.48 + 567.43 = \text{Tk. } 927.90$.
Check	The required return (12%) exceeds the coupon (10%), so the bond should sell below Tk. 1,000. It does.

BOND STATUS

The relation between price and face value is the **bond status**. A bond sells at a *discount* when its price is below face value, at *par* when equal, and at a *premium* when above. The status depends on the required return compared with the coupon rate.

REQUIRED RETURN AND BOND STATUS ($F = 1,000$; COUPON 10%; 5 YEARS)

Required return	Value (Tk.)	Status
12%	927.90	Discount
10%	1,000.00	Par
8%	1,079.85	Premium

Why does this happen? A buyer who can earn 12 per cent elsewhere will not pay Tk. 1,000 for a bond paying only 10 per cent. The price must fall until the bond too yields 12 per cent. When market rates fall, old bonds with higher coupons become more valuable. So bond prices move opposite to market interest rates.

Bond status Whether a bond sells at a discount, at par or at a premium.

When required return rises, bond price falls.

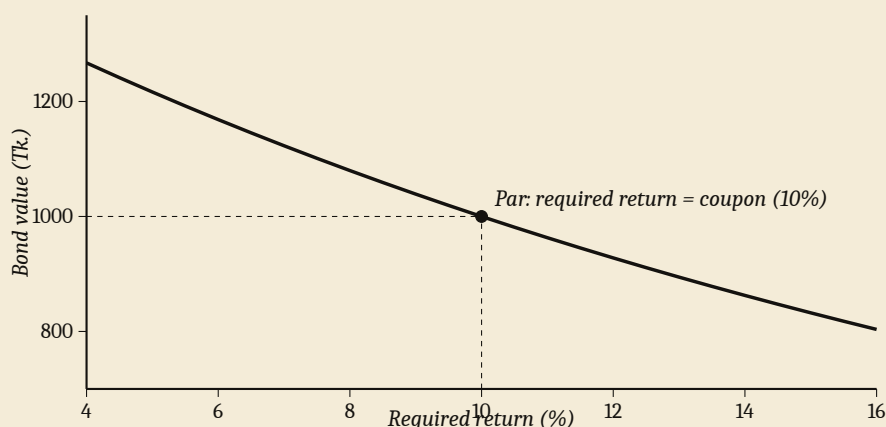


FIG. 7. Bond value falls as the required return rises.

SEMIANNUAL INTEREST AND YIELD TO MATURITY

Some bonds pay interest twice a year. We then halve the coupon, halve the rate, and double the periods.

WORKED CASE 2: SEMIANNUAL BOND	
Given	F = 1,000; coupon 8% paid twice yearly; n = 4 years; k = 10% a year.
Find	The value of the bond.
Formula	Use $I = \text{coupon} \div 2$, $k = \text{rate} \div 2$, periods = $2 \times \text{years}$.
Steps	$I = 40$ each half-year; $k = 5\%$; periods = 8 Annuity factor (5%, 8) = 6.4632; PV factor (5%, 8) = 0.6768 $40 \times 6.4632 = 258.53$; $1,000 \times 0.6768 = 676.84$
Answer	$B = 258.53 + 676.84 = \text{Tk. } 935.37$.
Check	The coupon (8%) is below the required return (10%), so the value is below 1,000. Correct.

Yield to maturity (YTM) The yearly return from buying a bond now and holding it to maturity.

The **yield to maturity (YTM)** is the yearly return a buyer earns if the bond is bought at today's price and held to maturity. It is the discount rate that makes the bond's value equal its price. A quick estimate is shown below.

$$YTM (\text{approximate}) = [I + (F - P) \div n] \div [(F + P) \div 2]$$

Discount price means YTM above the coupon rate.

For the bond priced at Tk. 927.90: $[100 + (1,000 - 927.90) \div 5] \div [(1,000 + 927.90) \div 2] = 114.42 \div 963.95 = 11.87$ per cent, close to the true 12 per cent. When the price is below face value, YTM is above the coupon rate; when above, YTM is below it.

VALUATION OF COMMON STOCK

Common stock Ordinary ownership shares with a vote and a claim on profit left over.

Go slowly here. Value comes from future dividends.

Go slowly here. **Common stock** means shares that give owners a vote and a claim on profit after all others are paid. A holder gains in two ways: dividends and a rise in price. But the price itself depends on future dividends, so we can value the stock by discount-

ing dividends alone.

Zero growth. Suppose a firm pays a steady dividend D every year, for ever. This is a perpetuity (Chapter 2), so value equals D divided by the required return k . If $D = \text{Tk. } 5$ and $k = 10$ per cent, the value is $5 \div 0.10 = \text{Tk. } 50$.

Constant growth. More often dividends grow at a steady rate g . Myron Gordon and Eli Shapiro described this case in 1956. Value is next year's dividend divided by the difference between k and g .

$$\text{Share value} = \text{next year's dividend} \div (\text{required return} - \text{growth rate})$$

WORKED CASE 3: CONSTANT GROWTH	
Given	$D_1 = \text{Tk. } 6$; $k = 13\%$; $g = 5\%$.
Find	The value of the share today.
Formula	$P_0 = D_1 \div (k - g)$
Steps	$k - g = 13\% - 5\% = 8\% = 0.08$ $P_0 = 6 \div 0.08$
Answer	$P_0 = \text{Tk. } 75$.
Check	If the share costs 75, next year's dividend yield is $6 \div 75 = 8\%$, and growth adds 5%: total 13% = k . It agrees.

EXCEPTIONS. The formula fails when g is greater than or equal to k , and it is sensitive: if g rises to 6%, value jumps to $6 \div 0.07 = \text{Tk. } 85.7$. Small errors in forecasts cause large errors in value. Use the result as a guide, not as a fact.

Small forecast errors cause large value errors.

BOND AND COMMON STOCK COMPARED

Feature	Bond	Common stock
Holder is	Lender	Owner
Return	Fixed interest	Dividends and price rise
Maturity	Fixed date	None
Priority in failure	Paid first	Paid last
Risk to holder	Lower	Higher
Vote	No	Yes

PRACTICE. (1) Value a bond: face Tk. 1,000, coupon 12%, 4 years, required return 10%. (2) Value a share with $D_1 = \text{Tk. } 3$, $k = 11\%$, $g = 5\%$.

Answers are at the back of the book.

SUMMARY. A security's value is the present value of its future cash at the required return. A bond combines an annuity and a single sum. Price moves opposite to market rates. A stock's value comes from dividends; constant growth gives $P_0 = D_1 \div (k - g)$.

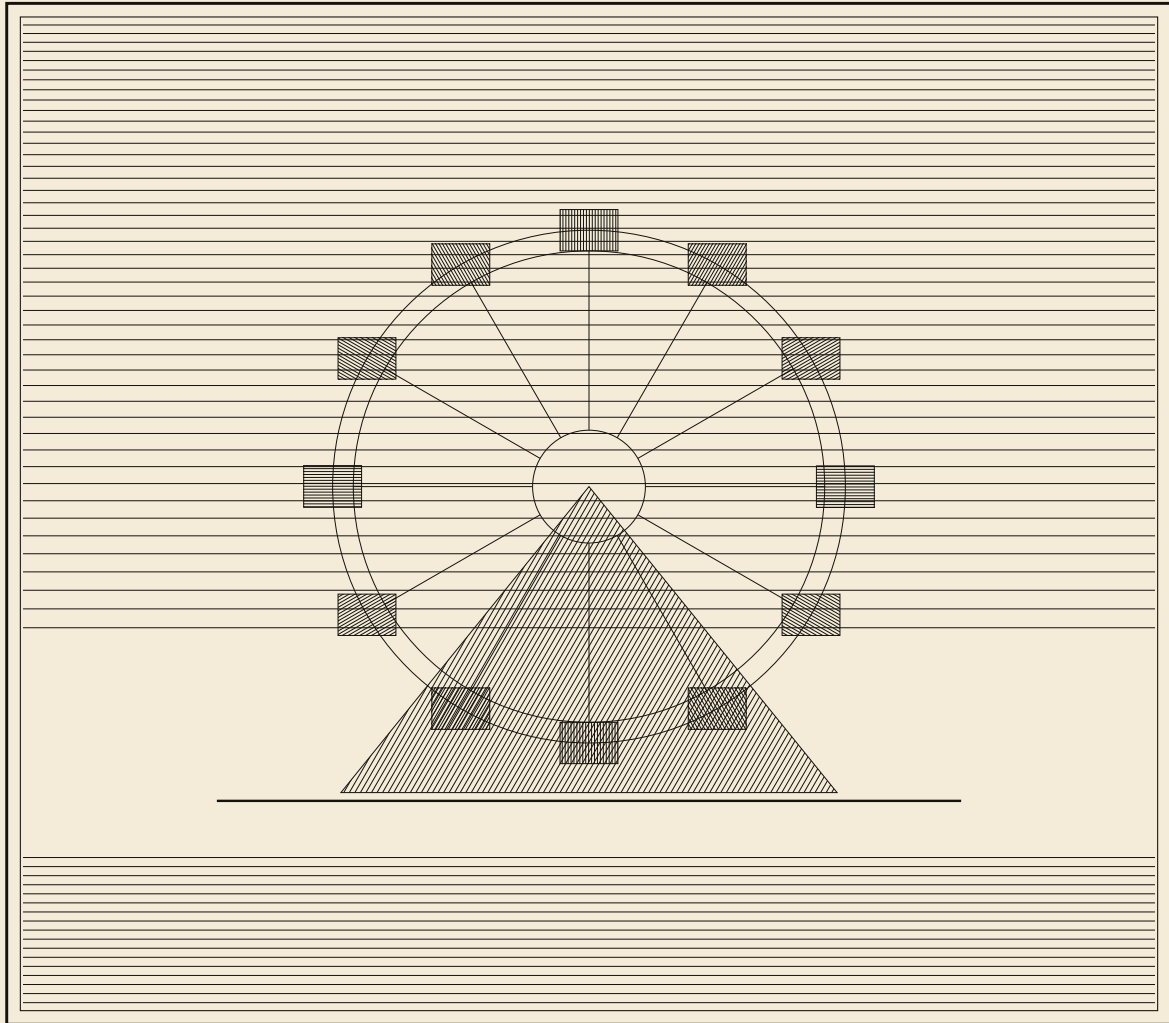
THINK IT THROUGH. A bond with $F = 1,000$, coupon 9%, 3 years, $k = 11\%$. Is it at a discount or premium? Estimate its value.

WHERE PEOPLE GO WRONG. Students use the coupon rate as the dis-

count rate; the discount rate is the required return. Others forget to halve the rate for semiannual bonds, or use g larger than k .

IN HISTORY. In 1751 the British government merged several debts into one stock called Consols, which paid fixed interest with no repayment date. It was a perpetuity, and in 1938 John Burr Williams, in *The Theory of Investment Value*, argued that a share is worth the present value of its dividends.

REVIEW QUESTIONS. (1) List four features of a bond. (2) When does a bond sell at a discount? (3) Find P_0 if $D_1 = 4$, $k = 12\%$, $g = 4\%$.



CHAPTER THE SIXTH

Managing Money Day by Day

How can a profitable firm run out of cash?



This chapter covers working capital: what it is, how it is financed in the short term, how to match financing to need, and how to estimate the amount a firm requires.

Profit and cash arrive at different times.

Working capital Money tied up in short-term assets such as cash, receivables and stock.

Net working capital Current assets minus current liabilities.

Operating cycle The time from paying for materials to collecting cash from customers.

Cash goes round: materials, goods, sales, cash.

Liquidity The ability to pay bills when they fall due.

Trade credit Buying goods now and paying the supplier later.

Accruals Costs already incurred but not yet paid, such as wages owed.

Cash discount A price reduction for paying early.

Karnaphuli Packaging sold boxes worth Tk. 40,00,000 in 2026 and made a profit. Yet in one week of September it could not pay its paper supplier, because customers had not yet paid. The firm was profitable and still short of cash. Why? Because profit is earned on the day of sale, but cash arrives only when the customer pays.

THE CONCEPT OF WORKING CAPITAL

Working capital means the money tied up in the firm's short-term assets: cash, customers' bills (*receivables*), and stock of materials and goods (*inventory*). These are *current assets*, which a firm expects to turn into cash within one year. *Current liabilities* are debts due within one year, such as unpaid supplier bills. **Net working capital** is current assets minus current liabilities.

Working capital moves in a circle called the **operating cycle**: cash buys materials; materials become goods; goods are sold on credit; customers pay, and cash returns. The longer the circle, the more money the firm needs to keep it turning.

Poor handling hurts in two ways. Too little working capital causes missed payments and lost sales. Too much leaves money idle that could earn a return elsewhere. The manager seeks a balance between *liquidity*, the ability to pay bills on time, and *profitability*.

SHORT-TERM FINANCING

Where does the money for working capital come from? The main sources are these.

Trade credit means buying goods now and paying later. Suppliers give it, and it is the largest source for small firms. **Accruals** are costs that a firm has incurred but not yet paid, such as wages owed at month end. A **bank loan** or overdraft allows the firm to borrow for short periods. *Factoring* means selling customers' bills to a company, called a factor, for immediate cash.

THE COST OF TRADE CREDIT

Trade credit looks free, but it is not always. A supplier may say *2/10, net 30*. This means a **cash discount** of 2 per cent if you pay within 10 days; otherwise pay the full amount in 30 days. Skipping the discount is a form of borrowing, and it has a cost.

$$\text{Cost of trade credit} = [\text{discount} \div (100 - \text{discount})] \times [365 \div (\text{final due day} - \text{discount day})]$$

WORKED CASE 1: COST OF GIVING UP A CASH DISCOUNT	
Given	Terms 2/15, net 45.
Find	The yearly cost of not taking the discount.
Formula	$\text{Cost} = [d \div (100 - d)] \times [365 \div (N - D)]$
Steps	$2 \div 98 = 0.02041$ $\text{Days gained} = 45 - 15 = 30; 365 \div 30 = 12.17$ $0.02041 \times 12.17 = 0.2483$
Answer	The cost is about 24.8% a year.
Check	If bank credit costs less than 24.8%, borrow and take the discount. Terms 2/10, net 30 would cost 37.2%.

THE MATCHING PRINCIPLE

Not all working capital is needed all year. **Permanent current assets** are the minimum level of stock and receivables that the firm holds all year. **Temporary current assets** are the extra amounts needed at busy seasons. The **matching principle** says: finance long-lasting needs with long-term funds, and temporary needs with short-term funds.

A firm that follows this principle takes a middle path. A firm that finances long needs with short loans is *aggressive*: its funds are cheaper, but it must keep renewing loans and may be refused. A firm that finances everything with long funds is *conservative*: it is safe, but pays for money it does not always need.

Matching principle Pay for long-lasting needs with long-term funds, and temporary needs with short-term funds.

Match the length of the loan to the length of the need.

Aggressive is cheaper but riskier; conservative is the reverse.

ESTIMATING WORKING CAPITAL NEEDS

The manager estimates how much to hold by asking how long each item stays in the firm. For each item, the amount equals yearly cost divided by 12, multiplied by the months held.

WORKED CASE 2: ESTIMATING WORKING CAPITAL	
Given	Annual sales 40,00,000; annual cost of production 30,00,000; materials bought 12,00,000. Raw material held 1 month; work in progress 0.5 month; finished goods 1 month; customers take 1.5 months to pay; supplier credit 1 month; minimum cash 50,000.
Find	Net working capital, with a 10% safety margin.
Formula	$\text{Item} = (\text{yearly cost} \div 12) \times \text{months held}$
Steps	Raw material: $1,00,000 \times 1 = 1,00,000$ Work in progress: $2,50,000 \times 0.5 = 1,25,000$ Finished goods: $2,50,000 \times 1 = 2,50,000$ Receivables (at cost): $2,50,000 \times 1.5 = 3,75,000$ Cash 50,000. Current assets = 9,00,000 Payables: $1,00,000 \times 1 = 1,00,000$ Net = 8,00,000; plus 10% = 8,80,000
Answer	Karnaphuli needs about Tk. 8,80,000.

Check	Current assets 9,00,000 less payables 1,00,000 gives 8,00,000. Margin 80,000 is 10%.
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Float Money tied up between writing a cheque and using the funds.

Float has three parts: mail, processing, clearing.

CASH AND FLOAT

Cheques take time to turn into usable cash. **Float** means the money tied up between the moment a cheque is written and the moment funds can be used. It has three parts: *mail float* (the cheque travels), *processing float* (the receiver records and deposits it), and *clearing float* (the bank clears it).

WORKED CASE 3: COST OF COLLECTION FLOAT	
Given	Daily receipts Tk. 50,000; mail 2 days; processing 1 day; clearing 3 days.
Find	Money tied up in float.
Formula	Money tied up = daily receipts × total float days
Steps	Total float = 2 + 1 + 3 = 6 days Tied up = 50,000 × 6
Answer	Tk. 3,00,000 is idle at any time.
Check	If the firm earns 10% on money, the yearly cost is Tk. 30,000. Faster collection saves it.

Compare cost, time and security for each source.

SHORT-TERM FINANCING IN BANGLADESH

A small firm in Bangladesh usually combines supplier credit with a bank overdraft or a short loan secured on stock or bills. Rules, limits and rates are set by banks and by Bangladesh Bank, and they change; check the current terms before planning. Whatever the source, compare cost, time to repay, and security required.

RECEIVABLES AND CREDIT POLICY

When a firm sells on credit, the money owed by customers becomes **receivables**. A firm chooses its **credit policy**: who may buy on credit, for how long, and what discount is offered for early payment. A generous policy increases sales, but it ties up more money and risks bad debts. A strict policy protects cash but may lose customers. The aim is the policy that adds most to value.

Receivables Money owed to the firm by customers who bought on credit.

Credit policy Rules on who may buy on credit and on what terms.

MANAGING INVENTORY

Inventory Stock of materials and goods that a firm holds.

Ordering cost The cost of placing and receiving one order.

Holding cost The yearly cost of keeping one unit in store.

Inventory means the stock of materials, partly finished goods and finished goods that a firm holds. Too much inventory ties up money and risks damage. Too little causes halts in production. Two costs pull in opposite directions. The **ordering cost** is the cost of placing one order. The **holding cost** is the yearly cost of keeping one unit in store. Large orders mean few orders but much stock;

small orders mean the reverse.

The **economic order quantity** (EOQ) is the order size that gives the lowest total of ordering and holding costs.

$$EOQ = \text{square root of } [(2 \times \text{yearly demand} \times \text{cost per order}) \div \text{holding cost per unit}]$$

WORKED CASE 4: ECONOMIC ORDER QUANTITY	
Given	Yearly demand $D = 12,000$ units; cost per order $S = \text{Tk. } 300$; holding cost $H = \text{Tk. } 6$ per unit a year.
Find	The EOQ and the yearly costs.
Formula	$EOQ = \sqrt{(2DS \div H)}$
Steps	$2 \times 12,000 \times 300 = 72,00,000$ $72,00,000 \div 6 = 12,00,000$ $\sqrt{12,00,000} = 1,095$ Orders $= 12,000 \div 1,095 = 10.96$; ordering cost $= 10.96 \times 300 = 3,288$ Holding cost $= (1,095 \div 2) \times 6 = 3,285$
Answer	EOQ = 1,095 units; total cost is about Tk. 6,573 a year.
Check	At the EOQ the two costs are nearly equal (3,288 and 3,285). This is the sign of the lowest total.

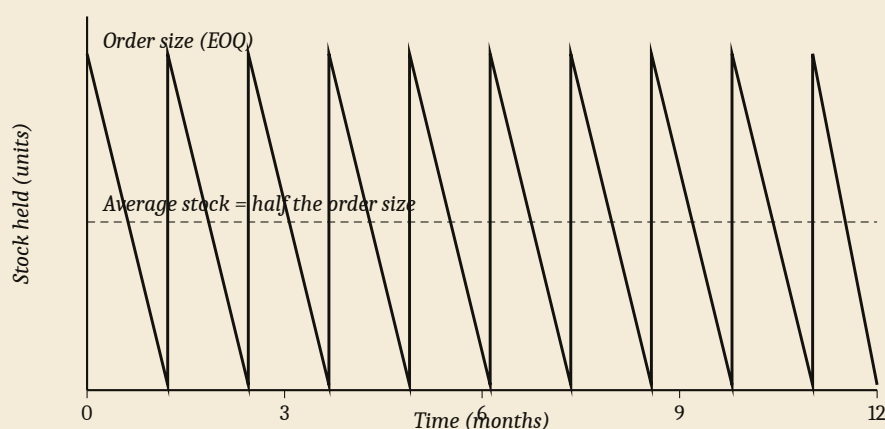


FIG. 8. Stock falls to zero, then is refilled by an order of EOQ units.

PRACTICE. (1) Find the cost of trade credit for terms 2/10, net 30. (2) Daily receipts are Tk. 40,000 and total float is 5 days. How much money is tied up?

SUMMARY. Working capital is the short-term money of the firm. Profit and cash differ in timing. Trade credit may cost more than it appears. The matching principle pairs short needs with short funds. Estimates multiply daily cost by days held.

THINK IT THROUGH. Compute the cost of trade credit for terms 3/10, net 30. Is a bank loan at 20% cheaper?

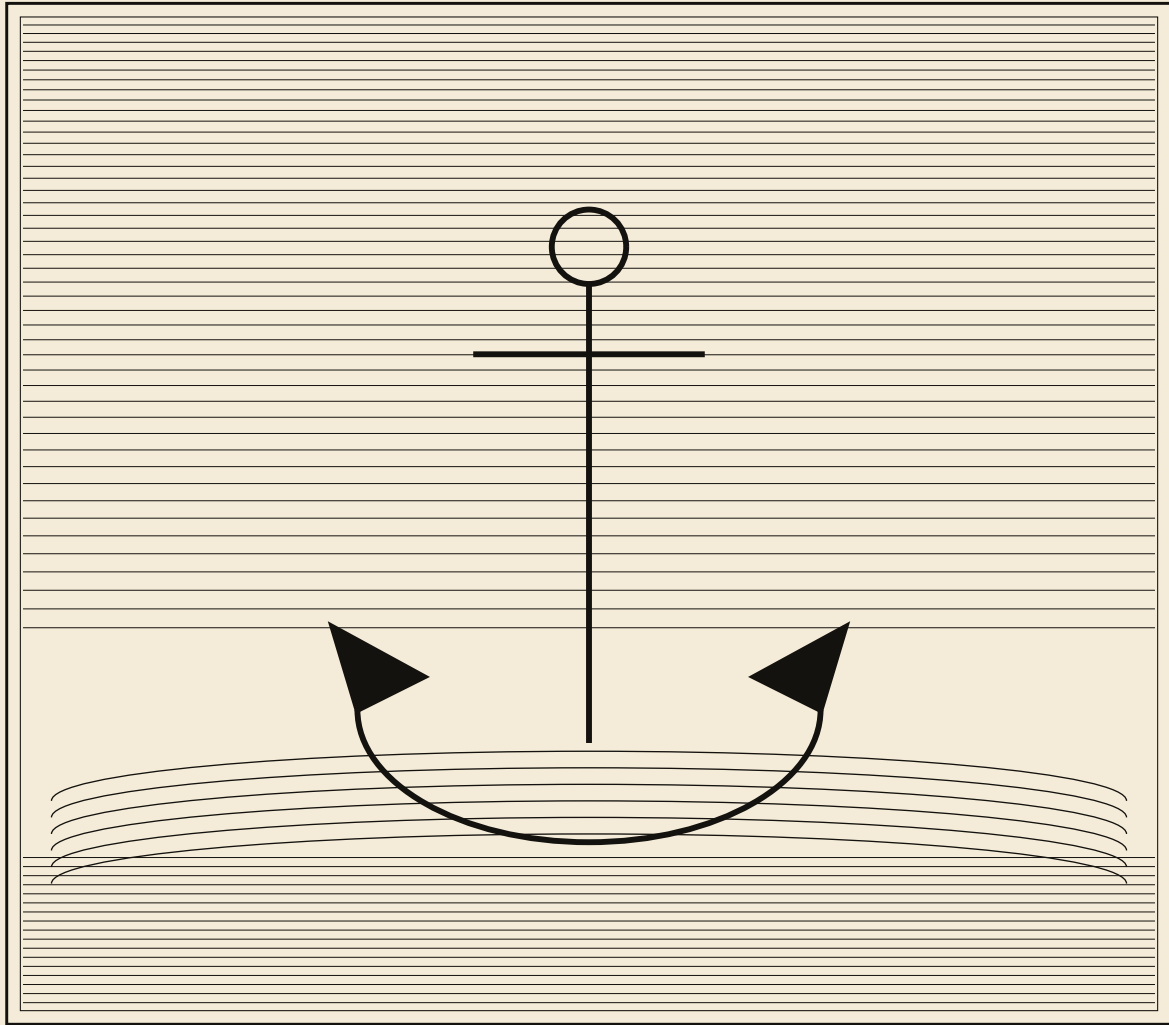
WHERE PEOPLE GO WRONG. Students call trade credit free. Others add receivables at sales value without noting the basis. They also mix up *net* working capital with current assets.

IN HISTORY. In 1952 William Baumol published a model that

Answers are at the back of the book.

treated cash like stock held in a store: holding too much costs interest, holding too little costs trips to the bank. In 1966 Merton Miller and Daniel Orr extended it to uneven cash flows.

REVIEW QUESTIONS. (1) Why can a profitable firm lack cash? (2) State the matching principle. (3) Find the cost of 1/10, net 40.



CHAPTER THE SEVENTH

Funding the Long Run

Where does a growing firm find money that it need not repay soon?



This chapter surveys long-term sources of funds: loans, bonds, preferred and common shares, and kept profit. It also shows how funds are raised in the capital market and which institutions supply them in Bangladesh.

*Long projects need long money;
the price varies by source.*

Short-term financing Money due for repayment within one year.

Intermediate-term financing Money due after one year and up to about five years.

Long-term financing Money due after more than five years, or never.

New firms need long funds while they build.

Judge each source by cost, time, claim, control, risk, tax.

Term loan A loan for a fixed period, repaid in agreed instalments.

All debt must be paid, profit or not.

In 2027 Karnaphuli plans a new factory hall costing Tk. 40,00,000. The firm has Tk. 8,00,000 in kept profit. The founders cannot pay the rest from their pockets, and a one-year loan would fall due before the hall earns anything. They need money that stays for many years. Where can they find it, and at what price?

SHORT, INTERMEDIATE AND LONG TERM

Financing is classified by the time before repayment. **Short-term financing** is due within one year; Chapter 6 described it. **Intermediate-term financing** (also called mid-term) is due in more than one year and up to about five years. **Long-term financing** is due after more than five years, or never. The limits are not exact and differ by lender.

Long-term funds matter most for a new business. A new firm must buy machines and premises that last many years. It has no record of profit yet, so it cannot lean on short loans that may not be renewed. Owners' money and long loans give it time to start earning.

GENERAL CHARACTERISTICS OF FINANCING

Every source can be judged on six points: its **cost** (what it must pay), its **maturity** (when it is repaid), the **claim** it gives on profit and assets, the **control** it gives the provider, its **risk** to the firm, and its **tax** effect. Interest on debt reduces taxable profit; dividends do not. These six points make comparison fair.

DEBT FINANCING

A **term loan** means a loan for a fixed period, repaid in agreed instalments, usually from a bank or specialised lender. It often needs *security*, meaning an asset such as land that the lender may take if the borrower fails to pay. A term loan may carry *covenants*, which are conditions the borrower must keep, such as limits on new debt.

Firms also borrow by issuing **bonds** (sometimes called *debentures* when no asset is pledged). A bond has fixed interest and a maturity date, as Chapter 5 explained. A firm may also acquire machines through a lease, taught in Chapter 8. These are the common *methods of debt*. All of them must be paid regardless of profit, and so they raise financial risk.

PREFERRED AND COMMON STOCK

Preferred stock means shares that pay a fixed dividend and rank before common shares for payment, but usually carry no vote. A firm may skip the dividend without default, though it must usually clear any missed amounts before paying common owners. Preferred stock mixes features of debt (fixed payment) and equity (no legal duty to pay).

Common stock is the main ownership of a firm. Its holders have limited liability, which means they cannot lose more than they paid for the shares. They have a vote, a claim on all profit left after others are paid, and often a *pre-emptive right*: the right to buy new shares first, so that their share of ownership is not diluted. Common stock has no due date and no required dividend, so it is the safest source for the firm and the riskiest for its holder.

The choice between debt and equity is a balance. Debt is cheaper after tax and keeps control with the founders. But it must be repaid, and too much of it raises financial risk (Chapter 4). Equity is safer for the firm but costs more and may share control. Most firms mix both, and the right mix depends on how steady their sales are.

Preferred stock Shares with a fixed dividend that rank before common shares.

Limited liability Owners cannot lose more than they paid for their shares.

Common stock: no due date, but owners take the risk.

Debt is cheap but risky; equity is safe but costly.

RETAINED EARNINGS

Profit kept in the firm, as Chapter 1 explained, is a source of long-term funds, too. It needs no issue costs and no new owners. But it is not free: owners could have received the money and invested it themselves. So retained earnings carry a cost equal to what owners expect to earn. A firm should keep profit only if it can invest it at least as well as owners could.

Kept profit has a cost: what owners could have earned.

LONG-TERM SOURCES COMPARED

Feature	Debt	Preferred	Common
Return to holder	Interest	Fixed dividend	Variable dividend
Due date	Yes	Usually none	None
Vote	No	Usually no	Yes
Tax effect	Interest saves tax	None	None
Risk to firm	High	Medium	Low

RAISING FUNDS FROM THE CAPITAL MARKET

The **capital market** means the market where long-term funds are raised and traded. In the *primary market* a firm sells new securities to investors for the first time and receives the money. In the

Capital market The market where long-term funds are raised and traded.

Primary market: firm sells new. Secondary: investors trade.

Initial public offering (IPO)

The first sale of a firm's shares to the public.

Rights issue An offer of new shares to current owners before anyone else.

Underwriter A financial firm that agrees to buy unsold shares for a fee.

Issue costs make new shares dearer than kept profit.

secondary market, such as a stock exchange, investors trade existing securities among themselves; the firm receives nothing but gains a daily price and easier future fundraising.

A firm that sells shares to the public for the first time makes an **initial public offering**, called an IPO. Other routes are a **rights issue**, which offers new shares to current owners first, and *private placement*, which sells to a few chosen investors. In Bangladesh the Bangladesh Securities and Exchange Commission (BSEC) regulates public issues, and shares trade on the Dhaka Stock Exchange (DSE) and the Chittagong Stock Exchange (CSE). Methods and rules change from time to time, so a firm must check the present rules before it acts.

A public issue usually involves an **underwriter**, which is a financial firm that agrees to buy any shares the public does not take. The firm pays a fee for this safety. Together with legal, printing and advertising costs, such **issue costs** reduce the money the firm receives. This is why the cost of new shares is higher than the cost of retained earnings.

INSTITUTIONS SUPPLYING LONG-TERM FINANCE IN BANGLADESH

MAIN INSTITUTIONS (CHECK PRESENT STATUS BEFORE USE)

Institution	Role
Commercial banks	Term loans; state-owned and private
Bangladesh Bank	Central bank; supervises banks, sets policy
Development banks	Long loans for industry (for example BDBL)
Investment Corporation of Bangladesh	Invests in and underwrites securities
Non-bank financial institutions	Term loans, leasing, deposits
Insurance companies	Invest long-term funds in bonds and shares
Stock exchanges and BSEC	Market and regulation of securities

THE COST OF EACH SOURCE

Debt is cheapest after tax; equity costs most.

Equity cost = dividend yield + growth.

Chapter 3 used the cost of capital. Here is how each part is found. The cost of **debt** after tax is the interest rate times $(1 - \text{tax rate})$; at 10 per cent and 30 per cent tax, it is 7 per cent. The cost of **preferred stock** is the fixed dividend divided by the net price received. A share paying Tk. 12 and sold at Tk. 100 less Tk. 2 for issue costs has a cost of $12 \div 98 = 12.24$ per cent.

The cost of **common stock** is the return owners expect. One way uses Chapter 5: $\text{cost} = D1 \div P0 + g$. With $D1 = \text{Tk. } 6$, $P0 = \text{Tk. } 75$

and $g = 5$ per cent, the cost is $8\% + 5\% = 13$ per cent. Another way uses CAPM from Chapter 4. Equity costs more than debt, because owners are paid last and bear more risk.

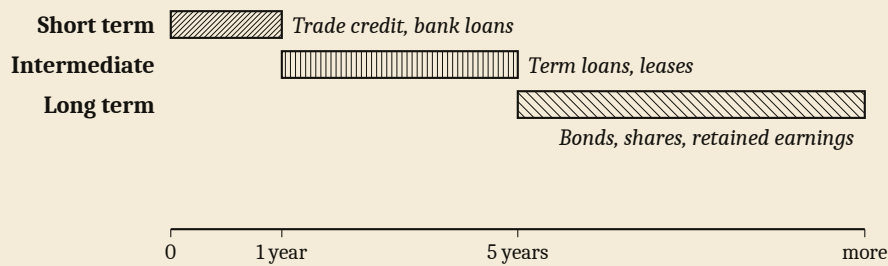


FIG. 9. Sources of finance arranged by the time before repayment.

WORKED CASE: CHOOSING A MIX	
Given	Karnaphuli seeks Tk. 40,00,000: retained earnings 8,00,000 (cost 14%), new common shares 12,00,000 (14%), bank term loan 20,00,000 (10% before tax). Tax 30%.
Find	The average cost of the Tk. 40,00,000.
Formula	Average = $\frac{\text{amount}}{\text{total}} \times \text{after-tax cost}$
Steps	Equity = 20,00,000 at 14%, weight 0.50 – 7.0% Debt = 20,00,000 at 7% after tax, weight 0.50 – 3.5% Sum = 7.0% + 3.5%
Answer	The average cost is 10.5%.
Check	It lies between 7% and 14%, as an average must. The firm would earn more than 10.5% from the hall for the plan to add value.

PRACTICE. Debt is 30% of funds at 12% before tax; equity is 70% at 16%; tax is 30%. Find WACC.

Before choosing a source, a manager asks five questions. How much money is needed, and for how long? Can the firm pay fixed amounts in a poor year? Will the owners accept sharing control? What does each source cost after tax and issue costs? Does the lender ask for security that the firm would rather keep free? The answers narrow the choice to a short list, and the cost of capital then ranks the list.

SUMMARY. Financing is short, intermediate or long term. Long-term sources are debt, preferred stock, common stock and retained earnings. Judge each by cost, maturity, claim, control, risk and tax. Public issues go through the capital market.

THINK IT THROUGH. A firm needs Tk. 40,00,000. It has 8,00,000 of retained earnings. Suggest a mix for the rest, and say which risk you accept.

WHERE PEOPLE GO WRONG. Students call retained earnings free, or think preferred stock is debt. Another error is to put the cost of debt before tax into WACC.

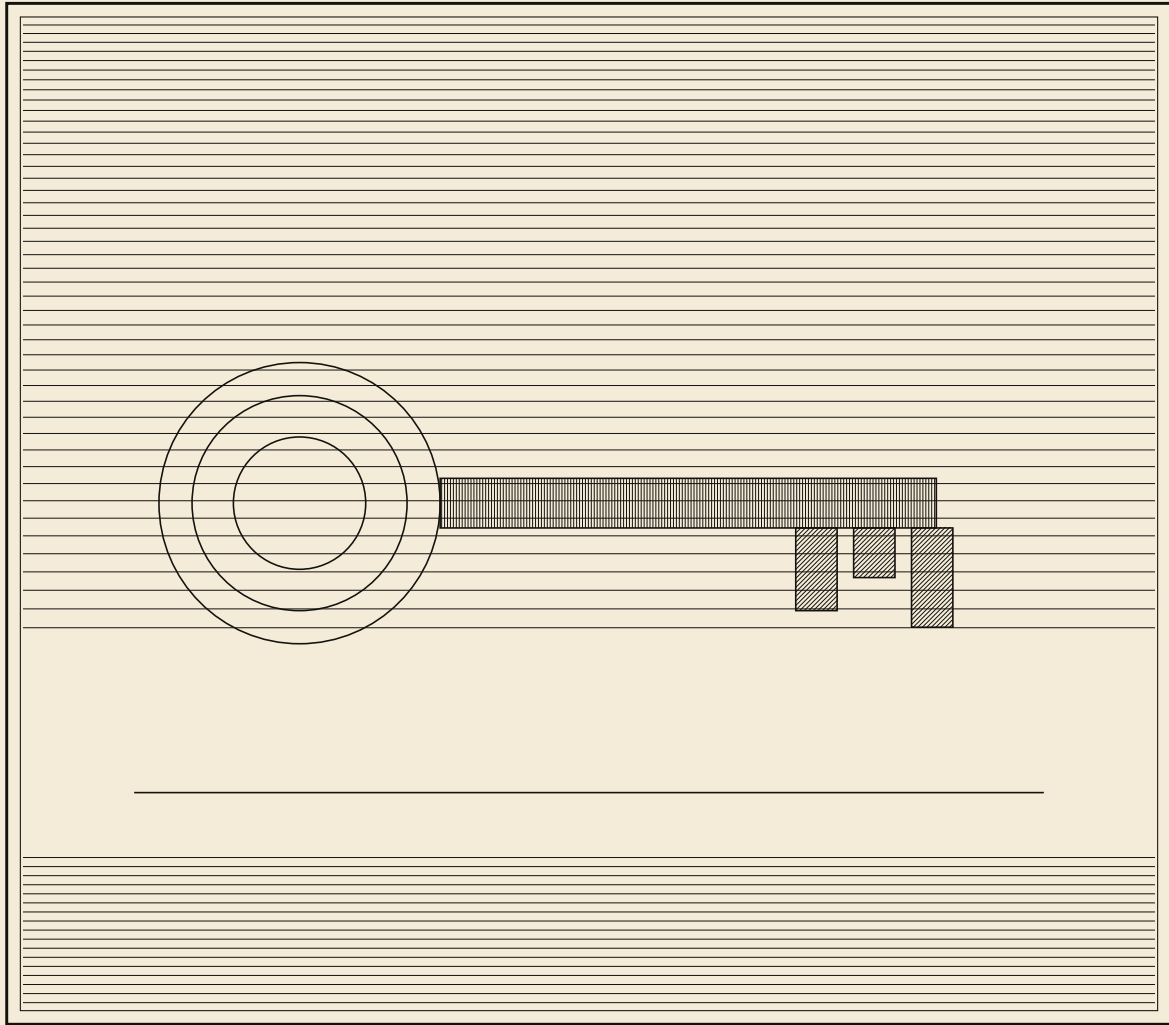
IN HISTORY. The stock exchange in Dhaka began in 1954 as the

Answers are at the back of the book.

Five questions narrow the choice of source.

East Pakistan Stock Exchange Association and took the name Dhaka Stock Exchange in 1964. It shows how a market can grow from a small group of brokers into a national institution.

REVIEW QUESTIONS. (1) Define intermediate-term financing. (2) Give two differences between preferred and common stock. (3) What is a rights issue?



CHAPTER THE EIGHTH

To Own or to Rent?

Must a firm own an asset in order to use it?



This chapter explains leasing, the two main kinds of lease, and the four-step method for deciding whether to lease an asset or buy it.

Use matters more than ownership.

A delivery van earns money by driving, not by being owned. Karnaphuli needs a van worth Tk. 5,00,000. It can buy the van with a bank loan, or it can pay a fee to someone who owns it. The van is identical in both cases. Only the way of paying differs. Which way costs the firm less?

Lease A contract allowing someone to use an asset for regular payments.

Lessor The owner who lets another use the asset.

Lessee The user who pays to use the asset.

Operating lease A short, cancellable lease; the lessor keeps ownership risks.

Financial lease A long, non-cancellable lease that works like a loan.

Leasing saves cash today but may cost more overall.

BASIC CONCEPTS

A **lease** means a contract in which the owner of an asset allows another party to use it for a fixed period in return for regular payments. The owner is the **lessor**; the user is the **lessee**. At the end, the asset normally returns to the lessor, unless the contract says otherwise.

TYPES OF LEASE

An **operating lease** is short compared with the asset's life. The lessor keeps the risks of owning and often services the asset, and the lessee may cancel with notice. A **financial lease** (or *finance lease*) covers most of the asset's life, cannot usually be cancelled, and the lessee carries the risks of owning. It resembles a loan.

OPERATING AND FINANCIAL LEASES COMPARED

Feature	Operating	Financial
Length	Short of asset's life	Most of asset's life
Cancel	Allowed with notice	Normally not allowed
Servicing	Often the lessor	The lessee
Risk of owning	Lessor	Lessee
Looks like	Renting	Borrowing to buy

In a *sale and leaseback*, a firm sells an asset it owns and leases it back, raising cash and keeping the use of the asset. Leasing helps firms with little cash, gives quick access to equipment, and may offer tax gains. It can cost more in total, and the firm never owns the asset.

GAINS AND LOSSES FROM LEASING

For the lessee	For the lessor
Little cash needed at start	Steady income from payments
Quick use of equipment	Keeps the asset as security
Less risk of becoming out of date	Gains from tax rules on ownership
May cost more in total	Risk that the lessee fails to pay

Lease payments may fall at the start or at the end of each period. Payments at the start form an annuity due (Chapter 2) and cost more in present-value terms.

WORKED CASE: LEASE PAYMENTS IN ADVANCE	
Given	Lease payment Tk. 1,00,000 a year for 3 years; discount rate 8%.
Find	The present value if paid (a) at year-end, (b) at the start of each year.
Formula	$PV = \text{payment} \times \text{annuity factor}$; annuity due = ordinary $\times (1 + i)$
Steps	Annuity factor (8%, 3) = 2.5771 (a) $1,00,000 \times 2.5771 = 2,57,710$ (b) $2,57,710 \times 1.08 = 2,78,327$
Answer	(a) Tk. 2,57,710; (b) Tk. 2,78,327.
Check	The due case is larger by 8%, because each payment is one year earlier.

THE LEASE-OR-BUY DECISION

The decision has four steps. First, find the after-tax cash outflows of **leasing**. Second, find the after-tax cash outflows of **purchasing** (here, with a loan). Third, discount each set at the after-tax cost of debt. Fourth, choose the plan with the lower present value of outflows. We discount at the cost of debt because both plans are fixed, loan-like payments.

Compare present values of after-tax outflows.

WORKED CASE: LEASE OR BUY A Tk. 5,00,000 VAN	
Given	Life 5 years; straight-line depreciation Tk. 1,00,000 a year; tax 30%. Lease: Tk. 1,40,000 at each year-end. Buy: 10% loan repaid by equal yearly instalments of Tk. 1,31,899. Discount rate: $10\% \times 0.70 = 7\%$.
Find	The cheaper plan in present-value terms.
Formula	After-tax outflow (buy) = instalment $- 0.30 \times (\text{interest} + \text{depreciation})$; (lease) = payment $\times 0.70$
Steps	Lease: $1,40,000 \times 0.70 = 98,000$ each year. $PV = 98,000 \times 4.1002 = 4,01,819$ Buy: interest by year = 50,000; 41,810; 32,801; 22,892; 11,991 Outflows = 86,899; 89,356; 92,058; 95,031; 98,302 Discounted at 7%, the sum is 3,76,994
Answer	Buying costs Tk. 3,76,994 in present value; leasing costs Tk. 4,01,819. Buying is cheaper by about Tk. 24,800.
Check	The lease payment, 1,40,000 a year, is larger than the loan instalment, 1,31,899, and gives no tax shield from depreciation. So leasing should cost more.

EXCEPTIONS. The answer changes with the tax rate, the discount rate, the resale value of the asset, and service costs. A lease may still be wise when cash is short or the asset becomes old quickly. Numbers are illustrative; use present rules and quotes.

Tax, resale value and need can reverse the result.

PRACTICE. A lease costs Tk. 1,50,000 a year for 4 years. Tax is 30% and the discount rate is 7%. Find the present value of after-tax lease payments.

Answers are at the back of the book.

SUMMARY. A lease lets a firm use an asset without owning it. Op-

erating leases are short and cancellable; financial leases are long and loan-like. The lease-or-buy decision compares the present value of after-tax outflows, discounted at the after-tax cost of debt.

THINK IT THROUGH. A lease payment is Tk. 1,20,000 a year for 4 years, with 30% tax. What is the after-tax payment, and the present value at 7%?

WHERE PEOPLE GO WRONG. Students forget the depreciation tax shield for purchase, discount at the wrong rate, or ignore tax on the lease payment.

IN HISTORY. Modern equipment leasing is often traced to 1952, when D. P. Boothe founded the United States Leasing Corporation in California. It showed that financing could be offered by a specialist who owned the machines and let firms use them.

REVIEW QUESTIONS. (1) Distinguish a lessor from a lessee. (2) Compare operating and financial leases. (3) List the four steps of lease-or-buy.

MATHS YOU WILL USE

This page is for reference. Come back to it whenever a calculation feels unfamiliar.

Percentages. A percentage is a number out of 100. To change a percentage into a decimal, divide by 100: $12\% = 0.12$. To find a percentage of a number, multiply: $12\% \text{ of Tk. } 5,000 = 0.12 \times 5,000 = \text{Tk. } 600$.

Percentage change. Percentage change = (new value – old value) $\div$ old value $\times$ 100. If sales rise from 80,000 to 88,000 cartons, the change is $8,000 \div 80,000 \times 100 = 10\%$.

Ratios. A ratio compares two numbers by division. Contribution of 16,00,000 and EBIT of 8,00,000 give a ratio of 2.0 (written 2 to 1).

Averages. The simple average adds the values and divides by how many there are. A weighted average multiplies each value by its weight first, as WACC does. The weights must add to 1 (or 100%).

Powers. $(1.10)^3$ means $1.10 \times 1.10 \times 1.10$. A negative power means division: $(1.10)^{-3} = 1 \div (1.10)^3$. Square root: $\sqrt{16} = 4$, because $4 \times 4 = 16$.

Rearranging an equation. Whatever you do to one side, do to the other. From $FV = PV \times F$ we get $PV = FV \div F$, and $F = FV \div PV$.

Rounding. Carry at least four decimal places in factors, and round only the final answer. Rounding early changes the last digits.

Reading a graph. The horizontal axis often shows time; the vertical axis shows money or return. A point shows the value at that time. A line that rises to the right shows growth.

The lakh. In Bangladesh numbers are grouped as 1,00,000 (one lakh) and 1,00,00,000 (one crore). Tables in this book use that grouping for taka and international grouping for pure factors.

A reference page, not a lesson.

Percent to decimal: divide by 100.

Change divided by the old value.

A ratio is one number divided by another.

Weighted average: value times weight, then add.

A power means repeated multiplication.

Do the same to both sides.

Round at the end, not in the middle.

Find the axes first, then the points.

TABLES OF FACTORS

n	Future value factor $(1 + i)^n$				Present value factor $1 \div (1 + i)^n$			
	8%	10%	12%	15%	8%	10%	12%	15%
1	1.0800	1.1000	1.1200	1.1500	0.9259	0.9091	0.8929	0.8696
2	1.1664	1.2100	1.2544	1.3225	0.8573	0.8264	0.7972	0.7561
3	1.2597	1.3310	1.4049	1.5209	0.7938	0.7513	0.7118	0.6575
4	1.3605	1.4641	1.5735	1.7490	0.7350	0.6830	0.6355	0.5718
5	1.4693	1.6105	1.7623	2.0114	0.6806	0.6209	0.5674	0.4972
6	1.5869	1.7716	1.9738	2.3131	0.6302	0.5645	0.5066	0.4323
7	1.7138	1.9487	2.2107	2.6600	0.5835	0.5132	0.4523	0.3759
8	1.8509	2.1436	2.4760	3.0590	0.5403	0.4665	0.4039	0.3269
9	1.9990	2.3579	2.7731	3.5179	0.5002	0.4241	0.3606	0.2843
10	2.1589	2.5937	3.1058	4.0456	0.4632	0.3855	0.3220	0.2472

n	Future value of annuity factor				Present value of annuity factor			
	8%	10%	12%	15%	8%	10%	12%	15%
1	1.0000	1.0000	1.0000	1.0000	0.9259	0.9091	0.8929	0.8696
2	2.0800	2.1000	2.1200	2.1500	1.7833	1.7355	1.6901	1.6257
3	3.2464	3.3100	3.3744	3.4725	2.5771	2.4869	2.4018	2.2832
4	4.5061	4.6410	4.7793	4.9934	3.3121	3.1699	3.0373	2.8550
5	5.8666	6.1051	6.3528	6.7424	3.9927	3.7908	3.6048	3.3522
6	7.3359	7.7156	8.1152	8.7537	4.6229	4.3553	4.1114	3.7845
7	8.9228	9.4872	10.0890	11.0668	5.2064	4.8684	4.5638	4.1604
8	10.6366	11.4359	12.2997	13.7268	5.7466	5.3349	4.9676	4.4873
9	12.4876	13.5795	14.7757	16.7858	6.2469	5.7590	5.3282	4.7716
10	14.4866	15.9374	17.5487	20.3037	6.7101	6.1446	5.6502	5.0188

Use: future value = present sum $\times$ FV factor; present value = future sum $\times$ PV factor. For other rates or periods, use the formulas in Chapter 2. Factors are rounded to four places, so answers may differ from calculator values in the last digits.

SHORT ANSWERS

CHAPTER 1. Review: (1) Finance is the study of how money is raised, used and managed over time. (2) Wealth maximisation counts time, risk and cash; profit does not. (3) Monitoring is a cost; pay linked to share price reduces it. Think: cutting repairs raises profit now, but breakdowns may lower value later.

CHAPTER 2. Review: (1) Money now is worth more than the same sum later, since it can earn. (2) An annuity due pays at the start of each period and is worth $(1 + i)$ times more. (3) $5,000 \div 1.21 = \text{Tk. } 4,132$. Think: $15,000 \times 3.0373 = \text{Tk. } 45,560$, so Tk. 50,000 now is better. Practice: (1) 88,117. (2) 43,553. (3) 1,05,519.

CHAPTER 3. Review: (1) The sums are large, long-lasting, hard to reverse and shape the firm. (2) It ignores cash after payback and ignores time value. (3) NPV = present value of inflows less cost; accept if above zero. Think: $2,00,000 \times 2.4869 = 4,97,370$; NPV = -2,630; reject. Practice: payback 3.0 years; NPV = $6,33,980 - 6,00,000 = \text{Tk. } 33,980$; accept.

CHAPTER 4. Review: (1) Business risk comes from operations; financial risk from debt. (2) $\text{DOL} = \text{contribution} \div \text{EBIT}$; $\text{DFL} = \text{EBIT} \div (\text{EBIT} - \text{interest})$; $\text{DTL} = \text{DOL} \times \text{DFL}$. (3) $\text{CV} = 4 \div 8 = 0.5$. Think: $\text{CV}(X) = 0.50$; $\text{CV}(Y) = 0.75$; Y is riskier per unit of return. More: (3) $E = 9.5\%$; $\text{SD} = 3.57\%$; $\text{CV} = 0.38$. (4) Plan 1: 17.5%; Plan 2: 28%. Practice: (1) contribution 3,00,000; EBIT 1,50,000; DOL 2.0; DFL 1.25; DTL 2.5. (2) $7\% + 0.8 \times 6\% = 11.8\%$.

CHAPTER 5. Review: (1) Face value, coupon rate, maturity, issuer, indenture. (2) When the required return is above the coupon rate. (3) $4 \div (0.12 - 0.04) = \text{Tk. } 50$. Think: discount; about Tk. 951. Practice: (1) Tk. 1,063.40. (2) Tk. 50.

CHAPTER 6. Review: (1) Profit is earned at sale; cash arrives when customers pay. (2) Finance long needs with long funds and short needs with short funds. (3) $(1 \div 99) \times (365 \div 30) = 12.3\%$. Think: 56.4%; a 20% loan is cheaper, so borrow and take the discount. Practice: (1) 37.2%. (2) Tk. 2,00,000.

CHAPTER 7. Review: (1) Due after one year, up to about five years. (2) Preferred: fixed dividend, usually no vote; common: variable dividend, vote. (3) An offer of new shares to current owners first. Think: for example, a long bank loan plus new common shares. Practice: $0.30 \times 8.4\% + 0.70 \times 16\% = 13.72\%$.

CHAPTER 8. Review: (1) The lessor owns and lets; the lessee uses and pays. (2) Operating: short, cancellable; financial: long, loan-like. (3) Find lease outflows; find purchase outflows; discount at after-tax cost of debt; choose the lower present value. Think: 84,000 after tax; $84,000 \times 3.3872 = \text{Tk. } 2,84,525$. Practice: $1,05,000 \times 3.3872 = \text{Tk. } 3,55,656$.

GLOSSARY

- Accounting.** Recording and reporting what a firm has done with its money. (Ch. 1)
- Accruals.** Costs already incurred but not yet paid, such as wages owed. (Ch. 6)
- Agency cost.** The cost of watching agents, rewarding them, and the loss that remains. (Ch. 1)
- Agency problem.** A clash between what agents want and what owners want. (Ch. 1)
- Agency relationship.** One person (the principal) hires another (the agent) to act for them. (Ch. 1)
- Annuity.** A series of equal payments at equal intervals for a fixed time. (Ch. 2)
- Annuity due.** Equal payments made at the start of each period. (Ch. 2)
- Annuity factor.** A number that turns equal payments into one total value. (Ch. 2)
- Beta.** A measure of how much a security moves with the whole market. (Ch. 4)
- Bond.** A loan to an issuer, repaid with fixed payments. (Ch. 5)
- Bond status.** Whether a bond sells at a discount, at par or at a premium. (Ch. 5)
- Business risk.** Uncertainty about operating income, caused by the business itself. (Ch. 4)
- Capital.** Money that owners and lenders put into a firm. (Ch. 1)
- Capital budgeting.** Planning and choosing long-term investments. (Ch. 3)
- Capital expenditure.** Money spent on assets that bring benefits for many years. (Ch. 3)
- Capital market.** The market where long-term funds are raised and traded. (Ch. 7)
- Capital rationing.** Ranking good projects and funding the best until money runs out. (Ch. 3)
- CAPM.** A model that sets required return from the risk-free rate, beta and the market premium. (Ch. 4)
- Cash discount.** A price reduction for paying early. (Ch. 6)
- Coefficient of variation.** Standard deviation divided by expected return: risk per unit of return. (Ch. 4)
- Common stock.** Ordinary ownership shares with a vote and a claim on profit left over. (Ch. 5)
- Compound interest.** Interest earned on both the original sum and earlier interest. (Ch. 2)
- Compounding.** Moving a sum forward in time by adding interest on interest. (Ch. 2)
- Conceptual framework.** The main ideas that tie a subject together. (Ch. 1)
- Contribution.** Sales minus variable cost. (Ch. 4)
- Corporate finance.** The branch of finance that serves firms' money decisions. (Ch. 1)
- Cost of capital.** The average rate a firm pays its lenders and owners for funds. (Ch. 3)
- Coupon rate.** Yearly interest on a bond, as a percentage of face value. (Ch. 5)
- Credit policy.** Rules on who may buy on credit and on what terms. (Ch. 6)
- Depreciation.** The share of an asset's cost charged as an expense each year. (Ch. 3)
- DFL.** Percentage change in earnings after interest for a one per cent change in EBIT. (Ch. 4)
- Discount rate.** The rate used to turn future sums into present values. (Ch. 2)
- Discounted payback period.** Time for discounted inflows to repay the cost. (Ch. 3)
- Discounting.** Moving a future sum back to its value today. (Ch. 2)
- Diversification.** Spreading money over many securities to reduce risk. (Ch. 4)
- Dividend decision.** Choosing how much profit to pay out and how much to keep. (Ch. 1)
- DOL.** Percentage change in EBIT for a one per cent change in sales. (Ch. 4)
- EBIT.** Earnings before interest and tax; the profit from operations. (Ch. 4)
- Effective annual rate.** The true yearly rate after compounding is counted. (Ch. 2)
- Efficient market.** A market where prices quickly reflect available information. (Ch. 4)
- Expected value.** The average of possible outcomes, weighted by their probabilities. (Ch. 4)
- Face value.** The amount a bond repays at maturity. (Ch. 5)
- Finance.** The study of raising, using and managing money over time. (Ch. 1)
- Financial lease.** A long, non-cancellable lease that works like a loan. (Ch. 8)
- Financial leverage.** Magnified profit change caused by fixed interest costs. (Ch. 4)
- Financial manager.** The person who makes or advises on a firm's money decisions. (Ch. 1)
- Financial planning.** Estimating how much money a firm will need, and when. (Ch. 1)
- Financial risk.** Extra uncertainty caused by the use of borrowed money. (Ch. 4)
- Financing decision.** Choosing where the firm's money will come from. (Ch. 1)
- Firm.** A business that sells goods or services to earn a profit. (Ch. 1)
- Float.** Money tied up between writing a cheque and using the funds. (Ch. 6)
- Future value.** What a sum today will be worth at a later date. (Ch. 2)
- Future value factor.** The number $(1 + i)^n$ that multiplies a present sum to give its future value. (Ch. 2)
- Holding cost.** The yearly cost of keeping one unit in store. (Ch. 6)
- Indenture.** The legal contract that sets out a bond's terms. (Ch. 5)
- Independent projects.** Projects that can all be accepted if each is good. (Ch. 3)
- Initial investment.** Cash paid at the start of a project. (Ch. 3)
- Initial public offering (IPO).** The first sale of a firm's shares to the public. (Ch. 7)
- Interest.** Payment for the use of someone else's money. (Ch. 2)
- Interest rate.** Interest as a percentage of the sum lent, for one period. (Ch. 2)
- Intermediate-term financing.** Money due after one year and up to about five years. (Ch. 7)
- Internal rate of return (IRR).** The discount rate at which a project's NPV is zero. (Ch. 3)
- Intrinsic value.** The worth of a security found by

- discounting its future cash. (Ch. 5)
- Inventory.** Stock of materials and goods that a firm holds. (Ch. 6)
- Investment decision.** Choosing which assets the firm should buy. (Ch. 1)
- Lease.** A contract allowing someone to use an asset for regular payments. (Ch. 8)
- Lessee.** The user who pays to use the asset. (Ch. 8)
- Lessor.** The owner who lets another use the asset. (Ch. 8)
- Leverage.** Using fixed costs to magnify the effect of sales changes on profit. (Ch. 4)
- Limited liability.** Owners cannot lose more than they paid for their shares. (Ch. 7)
- Liquidity.** The ability to pay bills when they fall due. (Ch. 6)
- Long-term financing.** Money due after more than five years, or never. (Ch. 7)
- Market risk.** Risk that affects all securities and cannot be diversified away. (Ch. 4)
- Matching principle.** Pay for long-lasting needs with long-term funds, and temporary needs with short-term funds. (Ch. 6)
- Maturity.** The date when a bond's face value is repaid. (Ch. 5)
- Mixed stream.** A series of cash flows of unequal amounts. (Ch. 2)
- Mutually exclusive projects.** Rival projects: accepting one means rejecting the others. (Ch. 3)
- Net present value (NPV).** Present value of a project's inflows minus its cost. (Ch. 3)
- Net working capital.** Current assets minus current liabilities. (Ch. 6)
- Nominal rate.** The yearly interest rate as quoted, before the effect of compounding. (Ch. 2)
- Operating cash flow.** Cash a project brings in each year of its life. (Ch. 3)
- Operating cycle.** The time from paying for materials to collecting cash from customers. (Ch. 6)
- Operating lease.** A short, cancellable lease; the lessor keeps ownership risks. (Ch. 8)
- Operating leverage.** Magnified profit change caused by fixed operating costs. (Ch. 4)
- Ordering cost.** The cost of placing and receiving one order. (Ch. 6)
- Ordinary annuity.** Equal payments made at the end of each period. (Ch. 2)
- Payback period.** Time needed for a project's inflows to repay its cost. (Ch. 3)
- Perpetuity.** An annuity that continues for ever. (Ch. 2)
- Portfolio.** A group of securities held together. (Ch. 4)
- Preferred stock.** Shares with a fixed dividend that rank before common shares. (Ch. 7)
- Present value.** What a future sum is worth today. (Ch. 2)
- Present value factor.** The number $1/(1+i)^n$ that turns a future sum into its present value. (Ch. 2)
- Probability distribution.** A list of possible outcomes with the chance of each. (Ch. 4)
- Profit maximisation.** Making accounting profit as large as possible. (Ch. 1)
- Profitability index.** Present value of inflows divided by the initial cost. (Ch. 3)
- Receivables.** Money owed to the firm by customers who bought on credit. (Ch. 6)
- Relevant cash flow.** Cash that changes if the project is accepted. (Ch. 3)
- Required return.** The minimum return an investor accepts, given the risk. (Ch. 4)
- Retained earnings.** Profit that the firm keeps instead of paying out. (Ch. 1)
- Return.** The gain or loss on an investment, as a percentage of its cost. (Ch. 4)
- Return on equity.** Profit after tax divided by the owners' money in the firm. (Ch. 4)
- Rights issue.** An offer of new shares to current owners before anyone else. (Ch. 7)
- Risk.** The chance that a result differs from what we expected. (Ch. 1)
- Risk averse.** Preferring the safer of two investments with the same expected return. (Ch. 4)
- Risk premium.** The extra return demanded for bearing risk. (Ch. 4)
- Risk-free rate.** The return on an asset with no chance of default. (Ch. 4)
- Risk-return trade-off.** Higher expected return goes with higher risk. (Ch. 4)
- Security.** A document giving its holder a claim on future cash. (Ch. 5)
- Share.** A part-ownership of a firm, with a claim on its profit. (Ch. 5)
- Short-term financing.** Money due for repayment within one year. (Ch. 7)
- Standard deviation.** A measure of how far outcomes spread around their average. (Ch. 4)
- Sunk cost.** Money already spent that cannot be recovered. (Ch. 3)
- Term loan.** A loan for a fixed period, repaid in agreed instalments. (Ch. 7)
- Terminal cash flow.** Cash received or paid when the project ends. (Ch. 3)
- Time value of money.** Money now is worth more than the same money later, because it can earn. (Ch. 2)
- Total leverage.** The combined effect of operating and financial leverage. (Ch. 4)
- Trade credit.** Buying goods now and paying the supplier later. (Ch. 6)
- Underwriter.** A financial firm that agrees to buy unsold shares for a fee. (Ch. 7)
- Variance.** Probability-weighted average of the squared gaps from the expected value. (Ch. 4)
- Wealth maximisation.** Making the value of the owners' shares as large as possible. (Ch. 1)
- Weighted average cost of capital (WACC).** The average cost of all funds, weighted by each source's share. (Ch. 3)
- Working capital.** Money tied up in short-term assets such as cash, receivables and stock. (Ch. 6)
- Yield to maturity (YTM).** The yearly return from buying a bond now and holding it to maturity. (Ch. 5)

FURTHER READING

Textbooks. Brigham and Houston, *Fundamentals of Financial Management*. Ross, Westerfield and Jordan, *Essentials of Corporate Finance*. Gitman and Zutter, *Principles of Managerial Finance*. Brealey, Myers and Allen, *Principles of Corporate Finance*. Van Horne and Wachowicz, *Fundamentals of Financial Management*. Editions change; use the latest one in your library.

Landmark works named in this book. Leonardo of Pisa (Fibonacci), *Liber Abaci* (1202). Irving Fisher, *The Theory of Interest* (1930). John Burr Williams, *The Theory of Investment Value* (1938). Joel Dean, *Capital Budgeting* (1951). Harry Markowitz, "Portfolio Selection", *Journal of Finance* (1952). William Baumol, "The Transactions Demand for Cash", *Quarterly Journal of Economics* (1952). Myron Gordon and Eli Shapiro, "Capital Equipment Analysis: The Required Rate of Profit", *Management Science* (1956). William Sharpe, "Capital Asset Prices", *Journal of Finance* (1964). Merton Miller and Daniel Orr, "A Model of the Demand for Money by Firms", *Quarterly Journal of Economics* (1966). Eugene Fama, "Efficient Capital Markets: A Review of Theory and Empirical Work", *Journal of Finance* (1970). Michael Jensen and William Meckling, "Theory of the Firm", *Journal of Financial Economics* (1976).

Bangladesh. For present rules on public issues, taxes and leasing, read the websites of Bangladesh Bank, the Bangladesh Securities and Exchange Commission, and the National Board of Revenue. Laws change; always check the current text.

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